



Intel® Xeon® 6 Processors: Birch Stream Platform

Performance and Power Optimization Guide

February 2026

Revision 1.8

Intel Confidential

Document Number: 819861

Notice: This document contains information on products in the design phase of development. The information here is subject to change without notice. Do not finalize a design with this information.

Intel technologies may require enabled hardware, software or service activation.

You may not use or facilitate the use of this document in connection with any infringement or other legal analysis concerning Intel products described herein. You agree to grant Intel a non-exclusive, royalty-free license to any patent claim thereafter drafted which includes subject matter disclosed herein.

No license (express or implied, by estoppel or otherwise) to any intellectual property rights is granted by this document.

The products described may contain design defects or errors known as errata which may cause the product to deviate from published specifications. Current characterized errata are available on request.

Performance results are based on testing as of dates shown in configurations and may not reflect all publicly available updates. See backup for configuration details. No product or component can be absolutely secure.

Performance varies by use, configuration, and other factors. Learn more on the [Performance Index site](#).

Your costs and results may vary.

"Conflict-free" refers to products, suppliers, supply chains, smelters, and refiners that, based on our due diligence, do not contain or source tantalum, tin, tungsten or gold (referred to as "conflict minerals" by the U.S. Securities and Exchange Commission) that directly or indirectly finance or benefit armed groups in the Democratic Republic of the Congo or adjoining countries.

All product plans and roadmaps are subject to change without notice.

Code names are used by Intel to identify products, technologies, or services that are in development and not publicly available. These are not "commercial" names and not intended to function as trademarks.

Intel disclaims all express and implied warranties, including without limitation, the implied warranties of merchantability, fitness for a particular purpose, and non-infringement, as well as any warranty arising from course of performance, course of dealing, or usage in trade.

Copies of documents which have an order number and are referenced in this document may be obtained by visiting the [Intel Resource and Document Center](#).

© 2026 Intel Corporation. Intel, the Intel logo, and other Intel marks are trademarks of Intel Corporation or its subsidiaries. Other names and brands may be claimed as the property of others.

Contents

1	Performance and Optimization Disclaimer	9
2	Preface	10
	2.1 Objectives	10
	2.2 Architecture & Performance Overview	10
	2.3 Additional Tuning Resources	11
3	Summary of Recommended Performance BIOS Options	13
	3.1 Workload: Intel® High Performance LINPACK	15
	3.2 Workload: SPEC CPU® 2017 Rate (Int and FP)	17
	3.3 Workload: STREAM Triad	18
	3.4 SPECpower_ssj® 2008	21
	3.5 SPECjbb® (Max) 2015	23
	3.6 SPECjbb® (Critical) 2015	25
	3.7 SPECchpc™ 2021	27
	3.8 HammerDB with MySQL	28
	3.9 MongoDB	30
	3.10 OLTP (SQL Server)	31
	3.11 OLAP (SQL Server)	32
	3.12 NGNIX	33
	3.13 VPP (Vector Packet Processor)	35
	3.14 AI Headnode	37
	3.15 IO Centric Workloads (Network and Storage)	38
	3.16 SAP BW Edition on SAP HANA	39
	3.17 MLPerf (AI Inference Benchmark)	40
A	Appendix	41
A	BIOS Descriptions	42
	A.1 Prefetchers	42
	A.1.1 Core Prefetchers	42
	A.1.2 Last Level Cache (LLC) Prefetch	43
	A.1.3 Homeless Prefetch	43
	A.1.4 AMP Prefetch	43
	A.1.5 Speculative Lock	44
	A.2 Uncore Frequency Ratio	44
	A.3 Priority Core Turbo (PCT)	44
	A.4 Snoop Mode	45
	A.5 Directory Mode Enable	46
	A.6 On-die Clustering Options	48
	A.6.1 2-way Sub-NUMA Clustering (SNC2)	49
	A.6.2 3-way Sub-NUMA Clustering (SNC3)	50
	A.6.3 VirtualNUMA Configuration	50
	A.7 TOR Thresholds (Loc2Rem)	51
	A.8 XPT Prefetch	53
	A.9 Dead Line LLC Allocation	53
	A.10 MSI Bypass	54
	A.11 Clock Modulation	54

A.12	Intel® Ultra Path Interconnect.....	55
A.12.1	Link Speed Mode.....	55
A.12.2	UPI Prefetch	55
A.12.3	Direct-to-UPI (D2U).....	55
A.13	Integrated I/O (IIO) Settings.....	56
A.13.1	Inbound Flows	57
A.13.2	Peer to Peer (P2P).....	63
A.13.3	Outbound Flows	64
A.13.4	Intel® DDIO Usage Tips.....	65
A.13.5	Common Buffer Direct Memory Access (CBDMA) Performance	65
A.13.6	Non-Transparent Bridge (NTB) Performance	65
A.13.7	Asynchronous Dynamic Refresh (ADR).....	66
A.14	Intel® Advanced Vector Extensions and Intel® Advanced Matrix Extensions.....	66
A.14.1	Recommended Performance Optimizations for Intel® Advanced Vector Extensions 2 (Intel® AVX2).....	66
A.14.2	Recommended Performance Optimizations for Intel® Advanced Vector Extensions 512 (Intel® AVX-512)	66
A.14.3	Recommended Performance Optimizations for Intel® Advanced Matrix Extensions (Intel® AMX)	67
A.15	Latency-Sensitive Recommendations	68
A.15.1	Energy/Performance Bias Recommendation for Latency-Sensitive Applications	68
A.15.2	Intel® Turbo Boost Technology Recommendation for Latency-Sensitive Applications	69
A.15.3	Enhanced Intel SpeedStep® Technology Recommendation for Latency-Sensitive Applications	70
A.15.4	BIOS Core Disable Recommendation for Latency-Sensitive Applications.....	71
A.15.5	Intel® Hyper-Threading (Intel HT) Technology Recommendation for Latency-Sensitive Applications	71
A.15.6	Restrictions to Micro-ops using *H instructions on Dispatch Ports	72
A.16	Benefits of Intel® Speed Select Technology (Intel® SST)	72
A.16.1	Intel® SST – Performance Profile (Intel SST-PP).....	73
A.16.2	Intel® SST – Base Frequency (Intel SST-BF).....	73
A.16.3	Intel® SST – Core Power (Intel SST-CP)	74
A.16.4	ActiveTurbo Frequency.....	74
B	Recommended Memory Configurations for Best Performance	75
B.1	Page Policy: (a.k.a. DRAM Page Mode).....	76
B.2	Non-Temporal Write Operations	77
C	Accelerator Performance Tuning	79
C.1	Intel® Advanced Matrix eXtensions (Intel® AMX)	79
C.2	Intel® QuickAssist Technology (Intel® QAT)	79
C.3	Intel® In-memory Analytics Accelerator (Intel® IAA).....	80
C.4	Intel® Data Streaming Accelerator (Intel® DSA).....	80
C.5	Intel® Dynamic Load Balancer (Intel® DLB)	81
C.6	Intel® Volume Management Device (Intel® VMD) and Intel® Virtual RAID on CPU (Intel® VROC).....	81

C.7	Intel® Software Guard Extensions (Intel® SGX).....	82
C.8	Compute Express Link™ (CXL).....	82
D	Recommendations for Energy Efficiency.....	83
D.1	Energy Efficiency Introduction.....	83
D.2	Suggested Memory Configuration	83
D.3	Workload Energy Efficiency Configurations	83
D.4	Networking Workload Energy Efficiency Considerations	87
E	Firmware and BIOS Settings Guidelines for Energy Efficiency	89
E.1	Power-Related BIOS Knobs.....	91
E.2	Hardware P-States (HWP)	91
E.3	Energy Perf Bias Cfg Mode.....	91
E.4	Dynamic Loadline Switch.....	92
E.5	ACPI C6x Enumeration	92
E.6	C1 to C1e Promotion.....	92
E.7	Uncore Freq Ratio (COMPUTE IO)	92
E.8	Uncore Freq Control (COMPUTE IO)	93
E.9	Latency Optimized Mode	93
E.10	Intel® Speed Shift Technology	93
F	Other Considerations	95
F.1	VR Design	95
F.2	Efficiency and Power Loss Equations	95
F.3	Elements that Contribute to Power Loss	95
F.4	MOSFET and Freewheeling Diode Conduction Losses.....	96
F.5	Printed Circuit Board (PCB) Losses	96
F.6	Inductor Losses	96
F.7	Controller and Driver Losses	97
F.8	Motherboard VR13.0 6 Phase Shedding in Low Power Mode	97
F.9	VCCIN MBVR IMON (Iout) Accuracy versus System Performance	97
F.10	System Thermal Design Considerations.....	98
F.11	Alternating Current (AC)/ Direct Current (DC) Power Supply Selection	99
F.12	USB Device Interactions.....	99
F.13	Processor Power Variation	101
Glossary	104	

Revision History

Revision Number	Description	Date
1.8	- Edit to SAP HANA Bios settings. - Changed XPT Prefetch from Enable to Disable for all platforms.	February 2026
1.7	Added MLPerf Inference optimization guidance	January 2026
1.6	- Updated links in References section - Updated Section 2.3 with new Workloads - BIOS tables in Section 3 were updated - Added section 3.14 for AI Headnode - Priority Core Turbo (PCT) added to Appendix A	December 2025
1.5	- Modified tables in Sections 3.10 and 3.11 - Modified WORKLOAD_CONFIG to WORKLD_CONFIG - Updated links in Reference section - Updated the tables to be searchable - Removed AMP Prefetch from Core Prefetchers	October 2025
1.4	- Section 3.14 renamed to "AI Headnode and IO Centric Workloads". It was "IO Centric Workload" - Table in Section 3.14 was expanded to include other BIOS knobs - Table in section 3.15 was edited for "XPT Prefetch" and "XPT local Prefetch" and "same as Enable" was added to those settings - Appendix A.7 was edited to better explain Auto, Disabled and Enabled setting - Table of Content was auto updated	October 2025
1.3	Added settings for SAP HANA Clean-up of BIOS knobs	February 2025
1.2	Added settings for OLTP (SQL Server), OLAP (SQL Server), VPP (Vector Packet Processor) and IO Centric Workload.	January 2025
1.1	Added more workloads and updated Xeon 6900P-series BIOS settings. Added notes to operating system section.	September 2024
1.0	Updates to descriptions and appendices added more workloads	June 2024
0.5	Initial release of the document.	April 2024
0.3	Initial draft release of the document for Intel® Xeon® 6 SP & AP processors in Birch Stream platforms	April 2024

References

Document	Location and Document Number
ENERGY STAR® Program Requirements for Computer Services, Version 2.1	https://www.energystar.gov/sites/default/files/asset/document/Version%202.1%20Computer%20Servers%20Program%20Requirements_1.pdf
Workstation and Server Platform Readiness Checks for the ENERGY STAR Program	Document number 411729: Contact your sales/field account team
Birch Stream BIOS Writer's Guide (BWG)	Document number 647898: https://cdrdv2.intel.com/v1/dl/getContent/647898
Sierra Forest Processor Architecture Specification	Document number 736624: https://www.intel.com/content/www/us/en/secure/content-details/736624/content-details.html
Sierra Forest Processor Registers Specification (formerly known as External Design Specification [EDS]: Volume 2, Registers)	Document number 737845: https://www.intel.com/content/www/us/en/secure/content-details/737845/content-details.html
Birch Stream-SP Platform Electrical Specification	Document number 645781: https://www.intel.com/content/www/us/en/secure/content-details/645781/content-details.html
Birch Stream-AP Platform Electrical Specification	Document number 635356: https://www.intel.com/content/www/us/en/secure/content-details/635356/content-details.html
Intel® Power Thermal Utility (Intel® PTU) for Windows*	Document number 637674: https://cdrdv2.intel.com/v1/dl/getContent/637674?explicitVersion=true
Intel® Power Thermal Utility (Intel® PTU) for Linux*	Document number 637673: https://cdrdv2.intel.com/v1/dl/getContent/637673?explicitVersion=true
Intel® Power Thermal Utility (Intel® PTU) User's Guide	Document number 615177: https://cdrdv2.intel.com/v1/dl/getContent/615177
Birch Stream-SP Platform Design Guide PDG	Document number 633502: https://cdrdv2.intel.com/v1/dl/getContent/633502
Birch Stream-AP Platform Design Guide PDG	Document number 635354: https://cdrdv2.intel.com/v1/dl/getContent/635354
Birch Stream Memory Configuration Guide	Document number 830694: https://www.intel.com/content/www/us/en/secure/content-details/830694/birch-stream-memory-configuration-guide.html?wapkw=Birch%20Stream%20Memory%20Configuration%20Guide
Intel® 64 and IA-32 Architectures Optimization Reference Manual	Document number 821612: https://www.intel.com/content/www/us/en/content-details/821612/intel-64-and-ia-32-architectures-optimization-reference-manual-volume-1.html?wapkw=Intel%C2%AE%2064%20and%20IA-32%20Architectures%20Optimization%20Reference%20Manual

Document	Location and Document Number
<i>Intel® QuickAssist Technology Developer Resources</i>	https://www.intel.com/content/www/us/en/developer/topic-technology/open/quick-assist-technology/overview.html
<i>Intel® QuickAssist Technology Performance Optimization Guide</i>	Document number 709209: https://cdrdv2.intel.com/v1/dl/getContent/709209
<i>Intel® Data Streaming Accelerator Architecture Specification</i>	Document number 671116: https://cdrdv2.intel.com/v1/dl/getContent/671116
<i>Intel® Data Streaming Accelerator User Guide</i>	Document number 759709 https://cdrdv2.intel.com/v1/dl/getContent/759709
<i>Intel® In-Memory Analytics Accelerator Architecture Specification</i>	Document number 721858: https://cdrdv2.intel.com/v1/dl/getContent/721858
<i>Intel® Dynamic Load Balancer 2.0 Programmer Guide</i>	Document number 613545: https://cdrdv2.intel.com/v1/dl/getContent/613545
<i>AI Tools, Libraries, and Framework Optimizations</i>	https://www.intel.com/content/www/us/en/developer/topic-technology/artificial-intelligence/tools.html
<i>Intel® 64 and IA-32 Architectures Software Developer Manuals</i>	https://software.intel.com/en-us/articles/intel-sdm?wapkw=intel%C3%82%C2%AE%2064%20and%20ia-32%20architectures%20optimization%20reference%20manual
<i>BIOS Settings for FlexRAN™ Reference Architecture Platforms Based on Intel® Xeon® Processors</i>	Document number 861549 : https://www.intel.com/content/www/us/en/secure/content-details/861549/bios-settings-for-flexran-reference-architecture-platforms-based-on-intel-xeon-processors-reference-guide.html?wapkw=BIOS%20Settings%20for%20FlexRAN%E2%84%A2%20Reference%20Architecture%20Platforms%20Based%20on%20Intel%C2%AE%20Xeon%C2%AE%20Processors
<i>BIOS Settings for Intel® Wireline, Cable, Wireless and Converged Access Platform</i>	Document number 747130: https://cdrdv2.intel.com/v1/dl/getContent/747130
<i>Priority Core Turbo (PCT) Technology Whitepaper</i>	https://cdrdv2.intel.com/v1/dl/getContent/846906

1 Performance and Optimization Disclaimer

Performance results are based on testing by Intel and may not reflect all publicly available security updates. See configuration disclosure for details. No product or component can be absolutely secure.

Intel does not control or audit third-party benchmark data or the web sites referenced in this document. You should visit the referenced web site and confirm whether referenced data is accurate.

Performance varies by use, configuration, and other factors. Learn more at www.intel.com/PerformanceIndex.

Optimization Notice: Intel's compilers may or may not optimize to the same degree for non-Intel microprocessors for optimizations that are not unique to Intel microprocessors. These optimizations include SSE2, SSE3, and SSSE3 instruction sets and other optimizations. Intel does not guarantee the availability, functionality, or effectiveness of any optimization on microprocessors not manufactured by Intel. Microprocessor-dependent optimizations in this product are intended for use with Intel microprocessors. Certain optimizations not specific to Intel microarchitecture are reserved for Intel microprocessors. Refer to the applicable product user and reference guides for more information regarding the specific instruction sets covered by this notice. Notice revision number: 20110804.

Intel® Advanced Vector Extensions (Intel® AVX) provides higher throughput to certain processor operations. Due to varying processor power characteristics, utilizing Intel® AVX instructions may cause a) some parts to operate at less than the rated frequency and b) some parts with Intel® Turbo Boost Technology 2.0 to not achieve any or maximum turbo frequencies. Performance varies depending on hardware, software, and system configuration and you can learn more at <http://www.intel.com/go/turbo>.

Intel® Hyper-Threading Technology (Intel® HT Technology) is available on select Intel processors. Requires an Intel HT Technology enabled system. Your performance varies depending on the specific hardware and software you use.

2 Preface

2.1 Objectives

This document contains guidelines that allow designers and manufacturers of Birch Stream-based platforms to tune them for best energy efficiency, to provide memory configuration and BIOS setting recommendations for the best out-of-the-box performance and scalability on the Intel® Xeon® 6 processor families-based servers. This includes recommended BIOS settings for two-socket (2S), four-socket (4S) and eight-socket (8S) based systems configured for best performance and energy efficiency.

This guide also contains sections on CPU, memory, firmware, and OS patches; as well as additional sections on validation tools, voltage regulator (VR) design, thermal considerations, and power supply selection. Each of these sections contains suggested guidelines only and may not be complete. The final implementation is up to the discretion of the platform designer or manufacturer.

2.2 Architecture & Performance Overview

The Intel Xeon 6 processor family (codenamed Sierra Forest and Granite Rapids) is the leadership Intel® Xeon® multi-core server processors. The AP processors support up to 52 bits of physical address space and 57 bits of virtual address space while the SP processors support up to 52 bits of physical address space and 48 bits of virtual address space and are designed for a platform consisting of at least one processor. Included in this family of processors are four Integrated Memory Controllers (IMC), Compute Express Link (CXL) 2.0, and an Integrated I/O (IIO) on a single silicon die. Most processor types support up to 96 lanes of PCI Express* 5.0 capable of 32GT/s, and 8 lanes of DMI while the R1S processor types support up to 136 lanes of PCI Express 5.0. Each IMC supports up to 2 channels of DDR5 DIMMs with up to 2 DIMMs per channel. The AP processors also support MCR DIMMs at 1 DIMM per channel.

The processor is comprised of multiple IO interface and CHA/core modules, and the modules are connected to a mesh interconnect which consists of horizontal and vertical interconnects, each interconnect consisting of bi-directional channels. Different SKUs of the processor may have a different combination of modules.

The Intel® Xeon® 6 processors with P-cores incorporate the Redwood Cove core, building upon the Golden Cove and Raptor Cove core utilized in 4th and 5th generations. The core implements a mid-level cache (MLC) of 2MB capacity.

The Intel® Xeon® 6 processors with E-cores incorporate the Sierra Glen core and deliver efficient performance for high-throughput workloads, ideal where power, space, and cooling are limited. Each e-core module has 4 cores that

share 4MB MLC and 3MB LLC. For an additional boost, built-in accelerators speed up performance for encryption, scale-out databases, and digital services. E-cores feature high memory bandwidth and rich I/O with up to 12 channels of DDR5, up to 64 lanes of CXL 2.0, and up to 96 lanes of PCIe 5.0. Intel Advanced Vector Extensions 2 (Intel AVX2) includes support for VNNI/INT8 and BF16/FP16.

The Input/Output (I/O) architecture has improved since the 5th gen microarchitecture. There are between 88 and 136 total lanes of PCI Express* (PCIe*) Gen5 vs. the 80 lanes of PCIe Gen5 available in the 4th Gen Intel® Xeon® family. See [A.13 Integrated I/O \(IIO\) Settings](#), for more details.

2.3 Additional Tuning Resources

Intel publishes tuning guides for a variety of server workloads. These guides are designed for people who are already familiar with the workload but are looking for system settings that will improve performance.

Tuning guides for databases, AI frameworks, HPC clusters, and network workloads systems can be found in the tables below, and online at this link: <https://software.intel.com/content/www/us/en/develop/articles/xeon-performance-tuning-and-solution-guides.html>.

Tuning Guides for Various Workloads
AI
BERT-based AI inference with Intel® AMX
Deep Learning Optimization
Databases
ClickHouse with Intel® IAA and Intel® AVX512
RocksDB Compression and Decompression with Intel® IAA
HPC
Intel® Xeon® CPU Max Series Tuning Guide
Media
Media Processing Basics
Networking
Open vSwitch with DPDK
User Space Network Stack Acceleration with Intel® DSA

Sample Intel® Xeon® Processors Workload and Software optimization recipes
Cassandra
Gluten
Java
Spark

[MySQL & PostgreSQL](#)

Optimized libraries and toolkits, developed by Intel, are available for AI, ML, and DL frameworks and numerous programming environments that can improve performance of various workloads. Here are links to more information:

Optimized Libraries and Tools
Intel Optimized AI Software
Intel Distribution for Python* , including NumPy, SciPy, scikit-learn*
Popular Deep Learning Frameworks including PyTorch, TensorFlow*, MXNet*,v PaddlePaddle*, Caffe*
Tools, libraries, and SDKs including Intel AI Analytics and Intel Distribution of OpenVINO* toolkits
oneAPI – a set of compilers, libraries, and porting, analysis and debugger tools.
Intel® Resource Director Technology (Intel® RDT) a framework for monitoring and allocation of shared cache and memory resources.

3 Summary of Recommended Performance BIOS Options

The following list of BIOS features are addressed in this section as they could have a direct impact on benchmark performance.

- Core Prefetchers (Data Cache Unit [DCU] IP, DCU Streamer, Mid-Level Cache [MLC] Spatial, MLC Streamer)
- Last Level Cache (LLC) prefetch
- Snoop Mode
- Directory Mode Enable
- Opportunistic Snoop Broadcast (OSB)
- Directory A-to-S (Stale AtoS or A2S)
- On-die Clustering Options with Uniform Memory Access (UMA) and Non-Uniform Memory Access (NUMA) clustering modes
- VirtualNUMA
- IO Directory Cache (IODC)
- XPT Prefetch
- KTI (UPI) Prefetch
- Deadline LLC allocation
- Direct-to-UPI (D2U)
- Intel® UPI Link Speed Mode
- Intel® Hyper-Threading technology (Intel® HT Technology)
- Intel® Turbo Boost Technology
- Hardware power management
- Energy/performance bias
- Intel® Speed Select Technology options
- Homeless Prefetch
- Loc2Rem thresholds Normal (Loc2Rem)
- Accelerator Performance Tuning
- UMA/NUMA Quadrant, Hemi, Hex, SNCx

New Performance knobs that have been introduced in the Birch Stream platform include:

- Uncore Frequency Control
 - Dynamic Loadline Switch
 - Workload Configuration
-

The following tables show the recommended BIOS settings to optimize the performance for each benchmark or class of benchmarks based on the tuning of the various Intel® Xeon® 6 processors.

Note: INTEL RECOMMENDS THAT CUSTOMERS EVALUATE THEIR RESPECTIVE WORKLOADS ON ALL SNOOP MODE CONFIGURATIONS FOR OPTIMAL PERFORMANCE.

These workload tables provide guidance for 1+socket platforms, with the understanding that all workloads are different, and the results may vary. The options suggested are not guaranteed to benefit any particular workload. These recommendations are made based on prior experience; not all combinations have been tested. The tables include those performance-related BIOS options that are recommended to be user-selectable. In the tables and the descriptions that follow the tables, the recommended BIOS knob setting is shown with an asterisk (*) preceding the default setting.

3.1 Workload: Intel® High Performance LINPACK

Category	BIOS knobs	Sierra Forest-SP	Granite Rapids-AP	Sierra Forest-AP	Granite Rapids-SP
Core	Enable LP [Global] Intel® Hyper-Threading Technology	N/A	Enable	N/A	Enable
	Intel® Turbo Boost Technology	Enable	Enable	Enable	Enable
Prefetchers	Core Prefetchers	All Enable	All Enable	All Enable	All Enable
	LLC Prefetch	Disable	Disable	Disable	Disable
	Homeless Prefetch	Auto	Auto	Auto	Auto
	AMP Prefetch	Enable	Enable	Enable	Enable
	XPT Prefetch	Auto [same as Enabled]	Auto [same as Enabled]	Auto [same as Enabled]	Auto [same as Enabled]
	DCU Streamer Prefetch	Disable	Enable	Auto	Auto
	DCU IP Prefetch	Enable	Enable	Enable	Enable
	Link L0p Enable	Auto	Auto	Auto	Auto
	Dead Line LLC alloc	Enable	Enable	Enable	Enable
	Loc2Rem	Auto	Auto	Auto	Auto
	Speculative Lock	Enable	Enable	Enable	Enable
Power Management	Hardware P-States	Native Mode	Native Mode	Native Mode	Native Mode
	CPU Power & Performance Policy	Performance	Performance	Performance	Performance
	Advanced Power Mgmt PM State Control	Native Mode	Native Mode	Native Mode	Native Mode with No Legacy Support
	latency optimized mode	Enable	Enable	Enable	Enable
	ACPI C6x Enumeration	Auto	Auto	Auto	Auto
	Enhanced Halt State (C1E)	Auto	Auto	Auto	Auto
Cluster Knobs	SNC/SNC2/SNC3	N/A	SNC3	SNC2	SNC2
	UMA-based Cluster	Enable	Disable	Disable	Disable
	Hemi/Quad/	HEMI	Hemisphere (2 clusters)	HEMI (SNC Auto)	Hemisphere (2 clusters)
	Virtual NUMA	Disable	Disable	Disable	Disable
UFS	Uncore Freq Control (aka Uncore Frequency Scaling)	Mode 1	Mode 1	Mode 1	Mode 1
Snoop Modes	Directory Mode Enable	Auto	Auto	Auto	Auto
	OSB Enabled	Auto	Auto	Auto	Auto
	IO Directory Cache (IODC)	Auto	Auto	Auto	Auto
	Stale AtoS	Auto	Auto	Auto	Auto
Intel® UPI	Direct to UPI (D2U)	Auto	Auto	Auto	Auto
	Page Policy	Closed Page	Closed Page	Closed Page	Closed Page
	Link Speed	Auto	Auto	Auto	Auto
	UPI Prefetch	Enable	Auto	Enable	Auto

Workload notes for Linux:

1. Use xhpl from the share/mkl/benchmarks/mp_linpack directory shipped with the latest oneMKL package. This will use a mixed MPI+OpenMP parallel runtime.
 2. Modify HPL.dat: P times Q must be equal to the number of NUMA domains.
 3. Nb should always be 384; N should be divisible by Nb, P and Q. N should be at least 220000. Higher N usually leads to better results, as long as the memory consumption is not larger than 80%.
 4. for single server with 2 CPUs and SNC=3 six NUMA domains are present. So, use P=3 and Q=2 ($P * Q = 3 * 2 = 6$); Nb=384; N=248832.
 5. rename runme_intel64_prv.txt to runme_intel64_prv
 6. rename runme_intel64_dynamic.txt to runme_intel64_dynamic
 7. make runme_intel64_dynamic and runme_intel64_prv executable
 8. modify runme_intel64_dynamic, setting MPI_PROC_NUM, MPI_PER_NODE and NUMA_PER_MPI. For example, for a single node with 2 CPUs and SNC3 ON (therefore 6 NUMA nodes, P=3, Q=2), use MPI_PROC_NUM=6, MPI_PER_NODE=6, NUMA_PER_MPI=1
 9. Source and Intel MPI environment file and execute runme_intel64_dynamic
-

3.2 Workload: SPEC CPU® 2017 Rate (Int and FP)

Category	BIOS knobs	Sierra Forest-SP	Granite Rapids-AP	Sierra Forest-AP	Granite Rapids-SP
Core	Enable LP [Global] Intel® Hyper-Threading Technology	N/A	ALL LPs	N/A	ALL LPs
	Intel® Turbo Boost Technology	Enable	Enable	Enable	Enable
Prefetchers	Core Prefetchers	All Enable	All Enable	All Enable	All Enable
	LLC Prefetch	Disable	Disable	Disable	Disable
	Homeless Prefetch	Auto	Auto	Auto	Auto
	AMP Prefetch	Enable	Enable	Enable	Enable
	XPT Prefetch	Auto [same as Enabled]	Auto [same as Enabled]	Auto [same as Enabled]	Auto [same as Enabled]
	DCU Streamer Prefetch	Disable	Disable	Auto	Auto
	DCU IP Prefetch	Enable	Enable	Enable	Enable
	Link L0p Enable	Auto	Auto	Auto	Auto
	Dead Line LLC alloc	Enable	Enable	Enable	Enable
	Loc2Rem	Auto	Auto	Auto	Auto
	Speculative Lock	Enable	Enable	Enable	Enable
Power Management	Hardware P-States	Native Mode	Native Mode	Native Mode	Native Mode
	CPU Power & Performance Policy	Performance	Performance	Performance	Performance
	Advanced Power Mgmt PM State Control	Native Mode	Native Mode	Native Mode	Native Mode with No Legacy Support
	latency optimized mode	Enable	Enable	Enable	Enable
	ACPI C6x Enumeration	Auto	Auto	Auto	Auto
	Enhanced Halt State (C1E)	Auto	Enable	Enable	Enable
Cluster Knobs	SNC/SNC2/SNC3	N/A	SNC3	SNC2	SNC2
	UMA-based Cluster	Enable	Disable	Disable	Disable
	Hemi/Quad/	HEMI	Hemisphere (2 clusters)	HEMI (SNC Auto)	Hemisphere (2 clusters)
	Virtual NUMA	Disable	Disable	Disable	Disable
UFS	Uncore Freq Control (aka Uncore Frequency Scaling)	Mode 1	Mode 1	Mode 1	Mode 1
Snoop Modes	Directory Mode Enable	Auto	Auto	Auto	Auto
	OSB Enabled	Auto	Auto	Auto	Auto
	IO Directory Cache (IODC)	Auto	Auto	Auto	Auto
	Stale AtoS	Auto	Auto	Auto	Auto
Intel® UPI	Direct to UPI (D2U)	Auto	Auto	Auto	Auto
	Page Policy	Auto	Auto	Auto	Auto
	Link Speed	Auto	Auto	Auto	Auto
	UPI Prefetch	Enable	Auto	Enable	Auto

Workload notes:

- GCC Compiler: -o2
- Intel Compiler: icx2024.0.2 or newer, binaries available from Intel

3.3 Workload: STREAM Triad

Category	BIOS knobs	Sierra Forest-SP	Granite Rapids-AP	Sierra Forest-AP	Granite Rapids-SP
Core	Enable LP [Global] Intel® Hyper-Threading Technology	N/A	Disable	N/A	Disable
	Intel® Turbo Boost Technology	Enable	Enable	Enable	Enable
Prefetchers	Core Prefetchers	All Enable	All Enable	All Enable	All Enable
	LLC Prefetch	Disable	Disable	Disable	Disable
	Homeless Prefetch	Auto	Auto	Auto	Auto
	AMP Prefetch	Enable	Enable	Enable	Enable
	XPT Prefetch	Auto [same as Enabled]	Auto [same as Enabled]	Auto [same as Enabled]	Auto [same as Enabled]
	DCU Streamer Prefetch	Disable	Enable	Auto	Auto
	DCU IP Prefetch	Enable	Enable	Enable	Enable
	Link L0p Enable	Auto	Auto	Auto	Auto
	Dead Line LLC alloc	Enable	Enable	Enable	Enable
	Loc2Rem	Auto	Auto	Auto	Auto
	Speculative Lock	Enable	Enable	Enable	Enable
Power Management	Hardware P-States	Native	Native	Native	Native
	CPU Power & Performance Policy	Performance	Performance	Performance	Performance
	Advanced Power Mgmt PM State Control	Native Mode	Native Mode	Native Mode	Native Mode with No Legacy Support
	latency optimized mode	Enable	Enable	Enable	Enable
	ACPI C6x Enumeration	Auto	Auto	Auto	Auto
	Enhanced Halt State (C1E)	Auto	Auto	Auto	Auto
Cluster Knobs	SNC/SNC2/SNC3	N/A	SNC3	SNC2	SNC2
	UMA-based Cluster	Enable	Disable	Disable	Disable
	Hemi/Quad/	HEMI	Hemisphere (2 clusters)	HEMI (SNC Auto)	Hemisphere (2 clusters)
	Virtual NUMA	Disable	Disable	Disable	Disable
UFS	Uncore Freq Control (aka Uncore Frequency Scaling)	Mode 1	Mode 1	Mode 1	Mode 1
Snoop Modes	Directory Mode Enable	Auto	Auto	Auto	Auto
	OSB Enabled	Auto	Auto	Auto	Auto

	IO Directory Cache (IODC)	Auto	Auto	Auto	Auto
	Stale AtoS	Auto	Auto	Auto	Auto
Intel® UPI	Direct to UPI (D2U)	Auto	Auto	Auto	Auto
	Page Policy	Closed Page	Closed Page	Closed Page	Closed Page
	Link Speed	Auto	Auto	Auto	Auto
	UPI Prefetch	Enable	Auto	Enable	Auto

Workload notes:

1. STREAM source code: <https://www.cs.virginia.edu/stream/FTP/Code/stream.c>
2. STREAM compilation macros: -DNTIMES=500 -DSTREAM_ARRAY_SIZE=<buffer size>
3. Intel Compiler, ICX options common for all Intel CPU's: -Wall -O3 -mcmodel=medium -qopenmp -fno-builtin -qopt-streaming-stores=always
4. Add the following additional ICX compiler option based on the supported instruction set architecture of Intel CPU:
 - a. For Intel Advanced Vector Extensions (Intel AVX): -xAVX
 - b. For Intel Advanced Vector Extensions 2 (Intel AVX2): -xCORE-AVX2
 - c. For Intel Advanced Vector Extensions 512 (Intel AVX-512): -xCORE-AVX512 -qopt-zmm-usage=high
5. Set the number of OpenMP threads to total number of available physical cores: export OMP_NUM_THREADS=<available_number_of_physical_cores>
6. Set OpenMP thread affinity:
 - a. If Hyper-Threads are enabled: export KMP_AFFINITY=granularity=fine,compact,1,0
 - b. If Hyper-Threads are disabled: export KMP_AFFINITY=compact
7. Number of array elements: The general rule for STREAM is that each array must be at least 4x the size of the sum of all the last-level caches used in the run, or 1 million elements -- whichever is larger. Refer to the table below to determine the array elements based on SKU type.

Sockets	Segment	CHAs	Array Elements (STREAM_ARRAY_SIZE)	Size of Array (Bytes)	Size of Array (MiB)	LLC Size (MiB)	4 x LLC	Meet Stream Rule (Size of Array >= 4 x LLC)
1	UCC	108	300,000,000	2,400,000,000	2288.8	432	1728	TRUE
1	UCC	120	300,000,000	2,400,000,000	2288.8	480	1920	TRUE
1	UCC	126	300,000,000	2,400,000,000	2288.8	504	2016	TRUE
2	UCC	108	600,000,000	4,800,000,000	4577.6	864	3456	TRUE
2	UCC	120	600,000,000	4,800,000,000	4577.6	960	3840	TRUE
2	UCC	126	600,000,000	4,800,000,000	4577.6	1008	4032	TRUE
1	XCC	84	200,000,000	1,600,000,000	1525.9	336	1344	TRUE
2	XCC	84	400,000,000	3,200,000,000	3051.8	672	2688	TRUE
1	XCC	72	200,000,000	1,600,000,000	1525.9	288	1152	TRUE
2	XCC	72	400,000,000	3,200,000,000	3051.8	576	2304	TRUE
1	HCC	48	120,000,000	960,000,000	915.5	192	768	TRUE
2	HCC	48	240,000,000	1,920,000,000	1831.1	384	1536	TRUE
1	LCC	18	50,000,000	400,000,000	381.5	72	288	TRUE
2	LCC	18	100,000,000	800,000,000	762.9	144	576	TRUE

8. Example of STREAM compilation command using Intel ICX Compiler on Intel Xeon 6 processors supporting AVX512 instructions (STREAM benchmark using 800 million array elements):

```
icx -Wall -O3 -mcmodel=medium -qopenmp -fno-builtin -qopt-streaming-stores=always -xCORE-AVX512 -qopt-zmm-usage=high -DNTIMES=500 -DSTREAM_ARRAY_SIZE=800000000 stream.c -o stream.bin
```

3.4 SPECpower_ssj® 2008

Category	BIOS knobs	Sierra Forest-SP	Granite Rapids-AP	Sierra Forest-AP	Granite Rapids-SP
Core	Enable LP [Global] Intel® Hyper-Threading Technology	N/A	ALL LPs	N/A	ALL LPs
	Intel® Turbo Boost Technology	Enable	Enable	Enable	Enable
Prefetchers	Core Prefetchers	All Enable	All Enable	All Enable	All Enable
	LLC Prefetch	Disable	Disable	Disable	Disable
	Homeless Prefetch	Auto	Auto	Auto	Auto
	AMP Prefetch	Enable	Enable	Enable	Enable
	XPT Prefetch	Auto [same as Enabled]	Auto [same as Enabled]	Auto [same as Enabled]	Auto [same as Enabled]
	DCU Streamer Prefetch	Disable	Disable	Auto	Auto
	DCU IP Prefetch	Enable	Enable	Enable	Enable
	Link L0p Enable	Auto	Auto	Auto	Auto
	Dead Line LLC alloc	Enable	Enable	Enable	Enable
	Loc2Rem	Auto	Auto	Auto	Auto
	Speculative Lock	Enable	Enable	Enable	Enable
Power Management	Hardware P-States	Disable	Disable	Native Mode	Disable
	CPU Power & Performance Policy	Performance	Performance	Performance	Performance
	Advanced Power Mgmt PM State Control	Native Mode	Native Mode	Native Mode	Native Mode with No Legacy Support
	Latency Optimized Mode	Enabled	Disable	Enable	Disable
	ACPI C6x Enumeration	Auto	Auto	Auto	Auto
	Enhanced Halt State (C1E)	Auto	Enable	Enable	Enable
Cluster Knobs	SNC/SNC2/SNC3	SNC2	SNC3	SNC2	SNC2
	UMA-based Cluster	Enable	Disable	Disable	Disable
	Hemi/Quad/	HEMI	Hemisphere (2 clusters)	HEMI (SNC Auto)	Hemisphere (2 clusters)
	Virtual NUMA	Enable	Enable	Disable	Enable
UFS	Uncore Freq Control (aka Uncore Frequency Scaling)	Mode 1	Mode 1	Mode 1	Mode 1
Snoop Modes	Directory Mode Enable	Auto	Auto	Auto	Auto
	OSB Enabled	Auto	Auto	Auto	Auto
	IO Directory Cache (IODC)	Auto	Auto	Auto	Auto
	Stale AtoS	Auto	Auto	Auto	Auto
Intel® UPI	Direct to UPI (D2U)	Auto	Auto	Auto	Auto
	Page Policy	Auto	Auto	Auto	Adaptive
	Link Speed	Auto	Auto	Auto	Auto
	UPI Prefetch	Enable	Auto	Enable	Auto

Workload notes:

- See Appendix E *Firmware and BIOS Settings Guidelines for Energy Efficiency* for complete details.
 - Turbo ON/OFF should be tested with each CPU to determine which setting produces the best overall score.
-

3.5 SPECjbb® (Max) 2015

Category	BIOS knobs	Sierra Forest-SP	Granite Rapids-AP	Sierra Forest-AP	Granite Rapids-SP
Core	Intel® HT Tech2 / Enable LP [Global]	N/A	ALL LPs	N/A	ALL LPs
	Intel® Turbo Boost Technology	Enable	Enable	Enable	Enable
Prefetchers	Core Prefetchers	All Enable	All Enable	All Enable	All Enable
	LLC Prefetch	Disable	Disable	Disable	Disable
	Homeless Prefetch	Auto	Auto	Auto	Auto
	AMP Prefetch	Enable	Enable	Enable	Enable
	XPT Prefetch	Auto [same as Enabled]	Auto [same as Enabled]	Auto [same as Enabled]	Auto [same as Enabled]
	DCU Streamer Prefetch	Disable	Enable	Auto	Auto
	DCU IP Prefetch	Enable	Enable	Enable	Enable
	Link L0p Enable	Auto	Auto	Auto	Auto
	Dead Line LLC alloc	Enable	Enable	Enable	Enable
	Loc2Rem	Auto	Auto	Auto	Auto
	Speculative Lock	Enable	Enable	Enable	Enable
Power Management	Hardware P-States	Native Mode	Native Mode	Native Mode	Native Mode
	CPU Power & Performance Policy	Performance	Performance	Performance	Performance
	Advanced Power Mgmt PM State Control	Native Mode	Native Mode	Native Mode	Native Mode with No Legacy Support
	latency optimized mode	Enable	Disable	Enable	Disable
	ACPI C6x Enumeration	Auto	Auto	Auto	Auto
	Enhanced Halt State (C1E)	Auto	Enable	Enable	Enable
Cluster Knobs	SNC/SNC2/SNC3	SNC2	Auto	SNC2	Auto
	UMA-based Cluster	Enable	Disable	Disable	Disable
	Hemi/Quad/	HEMI	Hemisphere (2 clusters)	HEMI	Hemisphere (2 clusters)
	Virtual NUMA	Disable	Disable	Disable	Disable
UFS	Uncore Freq Control (aka Uncore Frequency Scaling)	Mode 1	Mode 1	Mode 1	Mode 1
Snoop Modes	Directory Mode Enable	Auto	Auto	Auto	Auto
	OSB Enabled	Auto	Auto	Auto	Auto
	IO Directory Cache (IODC)	Auto	Auto	Auto	Auto
	Stale AtoS	Auto	Auto	Auto	Auto
Intel® UPI	Direct to UPI (D2U)	Auto	Auto	Auto	Auto
	Page Policy	Auto	Auto	Auto	Adaptive
	Link Speed	Auto	Auto	Auto	Auto
	UPI Prefetch	Enable	Auto	Enable	Auto

Workload notes:

- JDK 17 LTS, JDK 21 LTS, or later JDK
 - Utilize separate tuning/optimization SPECjbb*2015 guides available from Intel.
-

3.6 SPECjbb® (Critical) 2015

Category	BIOS knobs	Sierra Forest-SP	Granite Rapids-AP	Sierra Forest-AP	Granite Rapids-SP
Core	Enable LP [Global] Intel® Hyper-Threading Technology	N/A	ALL LPs	N/A	ALL LPs
	Intel® Turbo Boost Technology	Enable	Enable	Enable	Enable
Prefetchers	Core Prefetchers	All Enable	All Enable	All Enable	All Enable
	LLC Prefetch	Disable	Disable	Disable	Disable
	Homeless Prefetch	Auto	Auto	Auto	Auto
	AMP Prefetch	Enable	Enable	Enable	Enable
	XPT Prefetch	Auto [same as Enabled]	Auto [same as Enabled]	Auto [same as Enabled]	Auto [same as Enabled]
	DCU Streamer Prefetch	Enable	Enable	Auto	Auto
	DCU IP Prefetch	Enable	Enable	Enable	Enable
	Link L0p Enable	Auto	Auto	Auto	Auto
	Dead Line LLC alloc	Enable	Enable	Enable	Enable
	Loc2Rem	Auto	Auto	Auto	Auto
	Speculative Lock	Enable	Enable	Enable	Enable
Power Management	Hardware P-States	Native Mode	Native Mode	Native Mode	Native Mode
	CPU Power & Performance Policy	Performance	Performance	Performance	Performance
	Advanced Power Mgmt PM State Control	Native Mode	Native Mode	Native Mode	Native Mode with No Legacy Support
	latency optimized mode	Enable	Disable	Enable	Disable
	ACPI C6x Enumeration	Auto	Auto	Auto	Auto
	Enhanced Halt State (C1E)	Auto	Enable	Enable	Enable
Cluster Knobs	SNC/SNC2/SNC3	SNC2	Auto	SNC2	Auto
	UMA-based Cluster	Enable	Disable	Disable	Disable
	Hemi/Quad/	HEMI	Hemisphere (2 clusters)	HEMI (SNC Auto)	Hemisphere (2 clusters)
	Virtual NUMA	Disable	Disable	Disable	Disable
UFS	Uncore Freq Control (aka Uncore Frequency Scaling)	Mode 1	Mode 1	Mode 1	Mode 1
Snoop Modes	Directory Mode Enable	Auto	Auto	Auto	Auto
	OSB Enabled	Auto	Auto	Auto	Auto
	IO Directory Cache (IODC)	Auto	Auto	Auto	Auto
	Stale AtoS	Auto	Auto	Auto	Auto
Intel® UPI	Direct to UPI (D2U)	Auto	Auto	Auto	Auto
	Page Policy	Auto	Auto	Auto	Adaptive
	Link Speed	Auto	Auto	Auto	Auto
	UPI Prefetch	Enable	Auto	Enable	Auto

Workload notes:

- JDK 17 LTS, JDK 21 LTS, or later JDK
 - Utilize separate tuning/optimization SPECjbb*2015 guides available from Intel.
-

3.7 SPEChpc™ 2021

Category	BIOS knobs	Sierra Forest-SP	Granite Rapids-AP	Sierra Forest-AP	Granite Rapids-SP
Core	Enable LP [Global] Intel® Hyper-Threading Technology	N/A	Enable	N/A	Enable
	Intel® Turbo Boost Technology	Enable	Enable	Enable	Enable
Prefetchers	Core Prefetchers	All Enable	All Enable	All Enable	All Enable
	LLC Prefetch	Enable	Enable	Enable	Enable
	Homeless Prefetch	Enable	Enable	Enable	Enable
	AMP Prefetch	Enable	Enable	Enable	Enable
	XPT Prefetch	Auto [same as Enabled]	Auto [same as Enabled]	Auto [same as Enabled]	Auto [same as Enabled]
	DCU Streamer Prefetch	Enable	Enable	Enable	Enable
	DCU IP Prefetch	Enable	Enable	Enable	Enable
	Link L0p Enable	Auto	Auto	Auto	Auto
	Dead Line LLC alloc	Enable	Enable	Enable	Enable
	Loc2Rem	Auto	Auto	Auto	Auto
	Speculative Lock	Enable	Enable	Enable	Enable
Power Management	Hardware P-States	Native Mode	Native Mode	Native Mode	Native Mode
	CPU Power & Performance Policy	Performance	Performance	Performance	Performance
	Advanced Power Mgmt PM State Control	Native Mode	Native Mode	Native Mode	Native Mode with No Legacy Support
	latency optimized mode	Enable	Enable	Enable	Enable
	ACPI C6x Enumeration	Auto	Auto	Auto	Auto
	Enhanced Halt State (C1E)	Auto	Auto	Enable	Enable
Cluster Knobs	SNC/SNC2/SNC3	SNC3	SNC3	SNC3	SNC3
	UMA-based Cluster	Enable	Enable	Disable	Disable
	Hemi/Quad/	HEMI	Hemisphere (2 clusters)	HEMI (SNC Auto)	Hemisphere (2 clusters)
	Virtual NUMA	Disable	Disable	Disable	Disable
UFS	Uncore Freq Control (aka Uncore Frequency Scaling)	Mode 1	Mode 1	Mode 1	Mode 1
Snoop Modes	Directory Mode Enable	Auto	Auto	Auto	Auto
	OSB Enabled	Auto	Auto	Auto	Auto
	IO Directory Cache (IODC)	Auto	Auto	Auto	Auto
	Stale AtoS	Auto	Auto	Auto	Auto
Intel® UPI	Direct to UPI (D2U)	Auto	Auto	Auto	Auto
	Page Policy	Auto	Auto	Auto	Adaptive
	Link Speed	Auto	Auto	Auto	Auto
	UPI Prefetch	Enable	Auto	Enable	Auto

3.8 HammerDB with MySQL

Category	BIOS knobs	Sierra Forest-SP	Granite Rapids-AP	Sierra Forest-AP	Granite Rapids-SP
Core	Enable LP [Global] Intel® Hyper-Threading Technology	Default	Default	Default	Default
	Intel® Turbo Boost Technology	Default	Default	Default	Default
Prefetchers	Core Prefetchers	Default	Default	Default	Default
	LLC Prefetch	Default	Default	Default	Default
	Homeless Prefetch	Default	Default	Default	Default
	AMP Prefetch	Default	Default	Default	Default
	XPT Prefetch	Default	Default	Default	Default
	DCU Streamer Prefetch	Default	Default	Default	Default
	DCU IP Prefetch	Default	Default	Default	Default
	Link L0p Enable	Default	Default	Default	Default
	Dead Line LLC alloc	Default	Default	Default	Default
	Loc2Rem	Default	Default	Default	Default
	Speculative Lock	Default	Default	Default	Default
Power Management	Hardware P-States	Default	Default	Default	Default
	CPU Power & Performance Policy	Default	Default	Default	Default
	Advanced Power Mgmt PM State Control	Default	Default	Default	Default
	latency optimized mode	Default	Default	Default	Default
	ACPI C6x Enumeration	Default	Default	Default	Default
	Enhanced Halt State (C1E)	Default	Default	Default	Default
Cluster Knobs	SNC/SNC2/SNC3	Default	Default	Default	Default
	UMA-based Cluster	Default	Default	Default	Default
	Hemi/Quad/	Default	Default	Default	Default
	Virtual NUMA	Default	Default	Default	Default
UFS	Uncore Freq Control (aka Uncore Frequency Scaling)	Default	Default	Default	Default
Snoop Modes	Directory Mode Enable	Default	Default	Default	Default

	OSB Enabled	Default	Default	Default	Default
	IO Directory Cache (IODC)	Default	Default	Default	Default
	Stale AtoS	Default	Default	Default	Default
Intel® UPI	Direct to UPI (D2U)	Default	Default	Default	Default
	Page Policy	Default	Default	Default	Default
	Link Speed	Default	Default	Default	Default
	UPI Prefetch	Default	Auto	Enable	Auto

Workload notes:

- See HammerDB for additional workload parameters.

3.9 MongoDB

Category	BIOS knobs	Sierra Forest-SP	Granite Rapids-AP	Sierra Forest-AP	Granite Rapids-SP
Core	Enable LP [Global] Intel® Hyper-Threading Technology	N/A	ALL LPs	N/A	ALL LPs
	Intel® Turbo Boost Technology	Enable	Enable	Enable	Enable
Prefetchers	Core Prefetchers	All Enable	All Enable	All Enable	All Enable
	LLC Prefetch	Disable	Disable	Disable	Disable
	Homeless Prefetch	Auto	Auto	Auto	Auto
	AMP Prefetch	Enable	Enable	Enable	Enable
	XPT Prefetch	Auto [same as Enabled]	Auto [same as Enabled]	Auto [same as Enabled]	Auto [same as Enabled]
	DCU Streamer Prefetch	Disable	Enable	Auto	Auto
	DCU IP Prefetch	Enable	Enable	Enable	Enable
	Link L0p Enable	Auto	Auto	Auto	Auto
	Dead Line LLC alloc	Enable	Enable	Enable	Enable
	Loc2Rem	Auto	Auto	Auto	Auto
	Speculative Lock	Enable	Enable	Enable	Enable
Power Management	Hardware P-States	Native Mode	Native Mode	Native Mode	Native Mode
	CPU Power & Performance Policy	Performance	Performance	Performance	Performance
	Advanced Power Mgmt PM State Control	Native Mode	Native Mode	Native Mode	Native Mode with No Legacy Support
	latency optimized mode	Enable	Disable	Enable	Disable
	ACPI C6x Enumeration	Auto	Auto	Auto	Auto
	Enhanced Halt State (C1E)	Auto	Enable	Enabled	Enable
Cluster Knobs	SNC/SNC2/SNC3	SNC2	SNC3	SNC2	Auto
	UMA-based Cluster	Enable	Disable	Disable	Disable
	Hemi/Quad/	HEMI	Hemisphere (2 clusters)	HEMI (SNC Auto)	Hemisphere (2 clusters)
	Virtual NUMA	Disable	Disable	Disable	Disable
UFS	Uncore Freq Control (aka Uncore Frequency Scaling)	Mode 1	Mode 1	Mode 1	Mode 1
Snoop Modes	Directory Mode Enable	Auto	Auto	Auto	Auto
	OSB Enabled	Auto	Auto	Auto	Auto
	IO Directory Cache (IODC)	Auto	Auto	Auto	Auto
	Stale AtoS	Auto	Auto	Auto	Auto
Intel® UPI	Direct to UPI (D2U)	Auto	Auto	Auto	Auto
	Page Policy	Auto	Auto	Auto	Adaptive
	Link Speed	Auto	Auto	Auto	Auto
	UPI Prefetch	Enable	Auto	Enable	Auto

3.10 OLTP (SQL Server)

Category	BIOS knobs	Sierra Forest-SP	Granite Rapids-AP	Sierra Forest-AP	Granite Rapids-SP
Core	Enable LP [Global] Intel® Hyper-Threading Technology	N/A	Enable	N/A	Enable
	Intel® Turbo Boost Technology	Enable	Enable	Enable	Enable
Prefetchers	Hardware Prefetch	Disable	Disable	Disable	Disable
	Adjacent Cache Prefetch	Disable	Disable	Disable	Disable
	Homeless Prefetch	Enable	Enable	Enable	Enable
	AMP Prefetch	Disable	Disable	Disable	Disable
	XPT Prefetch	Disable	Disable	Disable	Disable
	DCU Streamer Prefetcher	Enable	Enable	Enable	Enable
	DCU IP Prefetch	Enable	Enable	Enable	Enable
	LLC Prefetch	Disable	Disable	Disable	Disable
	SLF Enable	Disable	Disable	Disable	Disable
Power Management	Hardware P-States	Disable	Disable	Disable	Disable
	Energy Efficient Turbo	Disable	Disable	Disable	Disable
	C1 to C1e Promotion	Disable	Disable	Disable	Disable
	Package C State	C0/C1 state	C0/C1 state	C0/C1 state	C0/C1 state
	Power Performance Tuning	BIOS Controls EPB	BIOS Controls EPB	BIOS Controls EPB	BIOS Controls EPB
	ENERGY_PERF_BIAS_CFG mode	Performance	Performance	Performance	Performance
	latency optimized mode	Enable	Enable	Enable	Enable
	Workload Configuration	I/O Sensitive	I/O Sensitive	I/O Sensitive	I/O Sensitive
	ACPI C6x Enumeration	Disable	Disable	Disable	Disable
	Enhanced Halt State (C1E)	Disable	Disable	Disable	Disable
Cluster Knobs	SNC/SNC2/SNC3	Enable	Enable	Enable	Enable
	Virtual NUMA	Enable	Enable	Enable	Enable
Snoop Modes	Directory Mode Enable	Default	Default	Default	Default
	OSB Enabled	Disable	Disable	Disable	Disable
	IO Directory Cache (IODC)	Default	Default	Default	Default
	Clock Modulation	Disable	Disable	Disable	Disable
	UPI L0p	Disable	Disable	Disable	Disable
	Stale AtoS	Enable	Enable	Enable	Enable
Intel® UPI	Direct to UPI (D2U)	Default	Default	Default	Default
	Page Policy	Closed	Closed	Closed	Closed
	Link Speed	Default	Default	Default	Default
	UPI Prefetch	Default	Default	Default	Default

3.11 OLAP (SQL Server)

Category	BIOS knobs	Sierra Forest-SP	Granite Rapids-AP	Sierra Forest-AP	Granite Rapids-SP
Core	Enable LP [Global] Intel® Hyper-Threading Technology	N/A	Enable	N/A	Enable
	Intel® Turbo Boost Technology	Enable	Enable	Enable	Enable
Prefetchers	Core Prefetch	Default	Default	Default	Default
	Homeless Prefetch	Default	Default	Default	Default
	AMP Prefetch	Disable	Disable	Disable	Disable
	XPT Prefetch	Default	Default	Default	Default
	LLC Prefetch	Default	Default	Default	Default
Power Management	Hardware P-States	Disable	Disable	Disable	Disable
	Energy Efficient Turbo	Disable	Disable	Disable	Disable
	C1 to C1e Promotion	Disable	Disable	Disable	Disable
	Package C State	C0/C1 state	C0/C1 state	C0/C1 state	C0/C1 state
	Power Performance Tuning	BIOS Controls EPB	BIOS Controls EPB	BIOS Controls EPB	BIOS Controls EPB
	ENERGY_PERF_BIAS_CFG mode	Performance	Performance	Performance	Performance
	latency optimized mode	Enable	Enable	Enable	Enable
	ACPI C6x Enumeration	Disable	Disable	Disable	Disable
	Enhanced Halt State (C1E)	Disable	Disable	Disable	Disable
Cluster Knobs	SNC/SNC2/SNC3	Enable	Enable	Enable	Enable
	UMA-based Cluster	Default	Default	Default	Default
	Virtual NUMA	Enable	Enable	Enable	Enable
Snoop Modes	Directory Mode Enable	Default	Default	Default	Default
	OSB Enabled	Default	Default	Default	Default
	IO Directory Cache (IODC)	Default	Default	Default	Default
	Stale AtoS	Default	Default	Default	Default
Intel® UPI	Direct to UPI (D2U)	Default	Default	Default	Default
	Page Policy	Adaptive	Adaptive	Adaptive	Adaptive
	Link Speed	Default	Default	Default	Default
	UPI Prefetch	Default	Default	Default	Default

3.12 NGNIX

Category	BIOS knobs	Sierra Forest-SP	Granite Rapids-AP	Sierra Forest-AP	Granite Rapids-SP
Core	Enable LP [Global] Intel® Hyper-Threading Technology	N/A	ALL LPs	N/A	ALL LPs
	Intel® Turbo Boost Technology	Disable	Disable	Disable	Disable
Prefetchers	Core Prefetchers	All Enable	All Enable	All Enable	All Enable
	LLC Prefetch	Disable	Disable	Disable	Disable
	Homeless Prefetch	Auto	Auto	Auto	Auto
	AMP Prefetch	Enable	Enable	Enable	Enable
	XPT Prefetch	Auto [same as Enabled]	Auto [same as Enabled]	Auto [same as Enabled]	Auto [same as Enabled]
	DCU Streamer Prefetch	Disable	Enable	Auto	Auto
	DCU IP Prefetch	Enable	Enable	Enable	Enable
	Link L0p Enable	Auto	Auto	Auto	Auto
	Dead Line LLC alloc	Enable	Enable	Enable	Enable
	Loc2Rem	Auto	Auto	Auto	Auto
	Speculative Lock	Enable	Enable	Enable	Enable
Power Management	Hardware P-States	Native Mode	Native Mode	Native Mode	Native Mode
	CPU Power & Performance Policy	Performance	Performance	Performance	Performance
	Advanced Power Mgmt PM State Control	Native Mode	Native Mode	Native Mode	Native Mode with No Legacy Support
	latency optimized mode	Enable	Disable	Enabled	Disable
	ACPI C6x Enumeration	Auto	Auto	Auto	Auto
	Enhanced Halt State (C1E)	Auto	Enable	Enable	Enable
Cluster Knobs	SNC/SNC2/SNC3	SNC2	SNC3	SNC2	Auto
	UMA-based Cluster	Enable	Disable	Disable	Disable
	Hemi/Quad/	HEMI	Hemisphere (2 clusters)	HEMI (SNC Auto)	Hemisphere (2 clusters)
	Virtual NUMA	Disable	Disable	Disable	Disable
UFS	Uncore Freq Control (aka Uncore Frequency Scaling)	Mode 1	Mode 1	Mode 1	Mode 1
Snoop Modes	Directory Mode Enable	Auto	Auto	Auto	Auto
	OSB Enabled	Auto	Auto	Auto	Auto
	IO Directory Cache (IODC)	Auto	Auto	Auto	Auto
	Stale AtoS	Auto	Auto	Auto	Auto
Intel® UPI	Direct to UPI (D2U)	Auto	Auto	Auto	Auto
	Page Policy	Auto	Auto	Auto	Adaptive
	Link Speed	Auto	Auto	Auto	Auto
	UPI Prefetch	Enable	Auto	Enable	Auto

Workload notes:

- Virtualization disabled
1. IIO Configuration/IOMMU: Disabled (Intel VT-D)
 2. Automatic NUMA: Disabled



3.13 VPP (Vector Packet Processor)

Category	BIOS knobs	Sierra Forest-SP	Granite Rapids-AP	Sierra Forest-AP	Granite Rapids-SP
Core	Enable LP [Global] Intel® Hyper-Threading Technology	N/A	All LPs	N/A	All LPs
	Intel® Turbo Boost Technology	Enable	Enable	Enable	Enable
Prefetchers	Core Prefetchers	All Enable	All Enable	All Enable	All Enable
	LLC Prefetch	Disable	Disable	Disable	Disable
	Homeless Prefetch	Auto	Auto	Auto	Auto
	AMP Prefetch	Enable	Enable	Enable	Enable
	XPT Prefetch	Auto	Auto	Auto	Auto
	DCU Streamer Prefetch	Disable	Enable	Auto	Auto
	DCU IP Prefetch	Disable	Disable	Disable	Disable
	Link L0p Enable	Auto	Auto	Auto	Auto
	Dead Line LLC alloc	Enable	Enable	Enable	Enable
	Loc2Rem	Auto	Auto	Auto	Auto
	Speculative Lock	Enable	Enable	Enable	Enable
Power Management	Hardware P-States	Native Mode	Native Mode	Native Mode	Native Mode
	CPU Power & Performance Policy	Performance	Performance	Performance	Performance
	Advanced Power Mgmt PM State Control	Native Mode	Native Mode	Native Mode	Native Mode with No Legacy Support
	latency optimized mode	Enable	Disable	Enable	Disable
	ACPI C6x Enumeration	Auto	Auto	Auto	Auto
	AVX License Pre-Grant Override	Enable	Enable	Enable	Enable
	AVX ICCP pre-grant level	512 Light	512 Light	512 Light	512 Light
	Enhanced Halt State (C1E)	Auto	Enable	Enable	Enable
Cluster Knobs	SNC/SNC2/SNC3	N/A	Enabled	Enabled	Enabled
	UMA-based Cluster	Enable	Disable	Disable	Disable
	Hemi/Quad/	SNC (Auto)	SNC (Auto)	SNC (Auto)	SNC (Auto)
	Virtual NUMA	Disable	Disable	Disable	Disable
Socket Configuration -> IIO Configuration	PCIe Snoop Holdoff Timer	0xB	0xB	0xB	0xB
	LLC IO Ways	0xfff	0xfff	0xfff	0xfff
	VT-d technology	Enable	Enable	Enable	Enable
	PCIe ASPM	Disable	Disable	Disable	Disable
UFS	Uncore Freq Control (aka Uncore Frequency Scaling)	Fixed Value	Fixed Value	Fixed Value	Fixed Value

Snoop Modes	Directory Mode Enable	Enable	Enable	Enable	Enable
	OSB Enabled	Auto	Auto	Auto	Auto
	IO Directory Cache (IODC)	Auto	Auto	Auto	Auto
	Stale AtoS	Auto	Auto	Auto	Auto
Intel® UPI	Direct to UPI (D2U)	Auto	Auto	Auto	Auto
	Page Policy	Auto	Auto	Auto	Adaptive
	Link Speed	Auto	Auto	Auto	Auto
	UPI Prefetch	Enable	Auto	Enable	Auto

Vector Packet Processor (VPP) workload notes (includes VPP FIB, IPsec, UPF, NGFW (<https://github.com/FDio/vpp>)):

- Benefits from distributing IO devices across IO Dies (NUMA node 0 and 1) for high throughput scenarios
- VT-d required for user-mode networking
- Uncore frequency value determined experimentally (trade off with power and core frequency)
- VPP variant setting = icl (enables AVX512 VPP node processing)
- DPDK plugin settings:
 - no-multi-seg
 - max-simd-bitwidth 512
- VPP Device queue sizes (descriptor rings) = 512 in tuned environments with low jitter. Increase as needed to achieve target packet loss ratio.

3.14 AI Headnode

AI Headnode usage requires single thread core and multi-port PCIe IO2Mem performance. If PCIe switch or proprietary switch (e.g. NVSwitch) is not present for accelerator attachment, good PCIe peer-to-peer performance is also desired.

Starting Xeon® 6 P-core Processor Family, selective SKUs offer Priority Core Turbo (PCT) feature to maximize single thread performance.

The following settings are recommended:

Category	BIOS knobs	Default Setting	BKM	Comments
Core	Enable LP [Global] Intel® Hyper-Threading Technology	All LPs	All LPs	
	Intel® Turbo Boost Technology	Enable	Enable	
	PCT Enable	Enable	Enable	
	PCT Partition Count	4 Partitions	Configurable	
	PCT Core Selection	0	1	
Prefetchers	Core Prefetchers	All Enable	All Enable	
	LLC Prefetch	Disable	Disable	
	Homeless Prefetch	Auto	Auto	
	AMP Prefetch	Enable	Enable	
	XPT Prefetch	Auto [same as Enabled]	Auto [same as Enabled]	
	DCU Streamer Prefetch	Disable	Enable	
	DCU IP Prefetch	Enable	Enable	
	Link L0p Enable	Auto	Auto	
	Speculative Lock	Enable	Enable	
	Dead Line LLC alloc	Enable	Enable	
Power Management	Hardware P-States	Disable	Disable	
	Package C state	Auto	C0/C1 state	
	CPU Power & Performance Policy	Performance	Performance	
	Package C State	C0/C1 state	C0/C1 state	
	Latency Optimized Mode	Disable	Enable	
	ACPI C6x Enumeration	Auto	Disable	
	Enhanced Halt State (C1E)	Enable	Disable	
	Workload Configuration	Balanced	I/O sensitive	
Uncore Configuration	SNC	Enable	Disable	
	Virtual NUMA	Disable	Disable	
UFS	Uncore Freq Control (aka Uncore Frequency Scaling)	Mode 1	Mode 1	
Snoop Modes	Directory Mode Enable	Auto	Auto	
	IO Directory Cache (IODC)	Auto	Auto	

Memory Configuration	Page Policy	Closed	Closed	Starting BKC#122 (Granite Rapids AP), BKC#110 (Granite Rapids SP), IFWI 2024.47.2.04 (BIOSID 3544.P24) onwards, Closed Page Mode becomes default.
IIO Configuration	PRS Capability for PCIe	Enable	Disable	If PRS (Page Request Service; an optimal extension of PCIe ATS protocol) is not required, the recommendation is to have this option disabled to avoid performance impact on small packets (8/16/32/64B) 1x16 configuration. (BHS SR#141).

3.15 IO Centric Workloads (Network and Storage)

The following settings are recommended:

Category	BIOS knobs	Default Setting	BKM	Comments
Core	Enable LP [Global] Intel® Hyper-Threading Technology	All LPs	All LPs	
	Intel® Turbo Boost Technology	Enable	Enable	
Prefetchers	Core Prefetchers	All Enable	All Enable	
	LLC Prefetch	Disable	Disable	
	Homeless Prefetch	Auto	Auto	
	AMP Prefetch	Enable	Enable	
	XPT Prefetch	Auto [same as Enabled]	Auto [same as Enabled]	
	DCU IP Prefetch	Enable	Enable	
	Link L0p Enable	Auto	Auto	
	Speculative Lock	Enable	Enable	
	Dead Line LLC alloc	Enable	Enable	
Power Management	Hardware P-States	Disable	Disable	
	Package C state	Auto	C0/C1 state	
	CPU Power & Performance Policy	Performance	Performance	
	Package C State	C0/C1 state	C0/C1 state	
	Latency Optimized Mode	Disable	Enable	
	C1 to C1e Promotion	Enable	Enable	
	ACPI C6x Enumeration	Auto	Disable	
	Enhanced Halt State (C1E)	Enable	Enable	
	Workload Configuration	Balanced	I/O sensitive	
Uncore Configuration	SNC	Enable	Enable	
UFS	Uncore Freq Control (aka Uncore Frequency Scaling)	Mode 1	Mode 1	
Common RefCode Configuration	Virtual NUMA	Disable	Disable	
Snoop Modes	Directory Mode Enable	Auto	Auto	
	IO Directory Cache (IODC)	Auto	Auto	
Memory Configuration	Page Policy	Closed	Closed	Starting BKC#122 (Granite Rapids AP), BKC#110 (Granite Rapids SP), IFWI 2024.47.2.04 (BIOSID 3544.P24) onwards, Closed Page Mode becomes default.

I/O Configuration	PRS Capability for PCIe	Enable	Disable	If PRS (Page Request Service; an optimal extension of PCIe ATS protocol) is not required, the recommendation is to have this option disabled to avoid performance impact on small packets (8/16/32/64B) 1x16 configuration. (BHS SR#141).
	Relax Ordering	Enable	Enable	
	Intel VT for Directed I/O (VT-d)->Intel® VT-d	Enable	Disable	
	PCIe ASPM	Per-Port	Disable	
	LTR Support	Auto	Auto	This feature is only applicable to certain network device modules.
	Toggle MSI Bypass	Disable(0x0)	Enable(0x1)	Refer to “17.1.5 MSI Bypass Performance Optimization” in Xeon 6 Tech Digest #850472
	MSI Bypass	Disable(0x0)	Enable(0x1)	
	PCIe 10-bit Tag Support	Auto(0x2)	Auto(0x2)	

3.16 SAP BW Edition on SAP HANA

Category	BIOS knobs	Granite Rapids 2S SP	Granite Rapids 8S SP	Granite Rapids 2S AP
Core	Enable LP [Global] Intel® Hyper-Threading Technology	Enable	Disable	Enable
Prefetchers	Core Prefetchers	All Enable	All Enable	All Enable
	LLC Prefetch	Disable	Disable	Disable
	Homeless Prefetch	Enable	Enable	Enable
	AMP Prefetch	Enable	Enable	Enable
	XPT Prefetch	Disable	Disable	Disable
	DCU Streamer Prefetch	Enable	Enable	Enable
	DCU IP Prefetch	Enable	Enable	Enable
Power Management	Hardware P-States	Disable	Disable	Disable
	CPU Power & Performance Policy	Performance	Performance	Performance
	Advanced Power Mgmt PM State Control	C0/C1	C0/C1	C0/C1
	Latency Optimized Mode	Enable	Enable	Enable
	ACPI C6x Enumeration		Enable	
	Enhanced Halt State (C1E)	Enable	Enable	Enable
Cluster Knobs	SNC/SNC2/SNC3	SNC2	SNC2	SNC3
Snoop Modes	Directory Mode Enable		Auto	
	OSB Enabled	Enable	Disable	Enable
	Stale AtoS	Enable	Enable	Enable
	Page Policy	Adaptive	Adaptive	Adaptive
	UPI Prefetch		Enable	

3.17 MLPerf (AI Inference Benchmark)

Category	BIOS knobs	Granite Rapids-AP	Granite Rapids-SP
Core	Enable LP [Global] Intel® Hyper-Threading Technology	ALL LPs	ALL LPs
	Intel® Turbo Boost Technology	Enable	Enable
Prefetchers	Core Prefetchers	All Enable	All Enable
	LLC Prefetch	Disable	Disable
	Homeless Prefetch	Enable	Enable
	AMP Prefetch	Enable	Enable
	XPT Prefetch	Auto	Auto
	DCU Streamer Prefetch	Enable	Enable
	DCU IP Prefetch	Enable	Enable
	Link L0p Enable	Auto	Auto
	Dead Line LLC alloc	Enable	Enable
	Loc2Rem	Auto	Auto
	L1 Next Page Prefetcher	Enable	Enable
Power Management	Hardware P-States	Native	Native
	CPU Power & Performance Policy	Performance	Performance
	Energy Efficient Turbo	Disable	Disable
	C1 to C1e Promotion	Disable	Disable
	ACPI C6x Enumeration	Disable	Disable
	Power Performance Tuning	OS Control EPB	OS Control EPB
	ENERGY_PERF_BIAS_CFG mode	Performance	Performance
	Latency optimized node	Enable	Enable
	Enhanced Halt State (C1E)	Disable	Disable
Cluster Knobs	SNC/SNC2/SNC3	Enable	Enable
	UMA-based Cluster	Disable	Disable
	Virtual NUMA	Disable	Disable
Snoop Modes	Directory Mode Enable	Auto	Auto
	OSB Enabled	Auto	Auto
	IO Directory Cache (IODC)	Auto	Auto
	Stale AtoS	Auto	Auto
	Page Policy	Auto	Auto
	UPI Prefetch	Auto	Auto

A *Appendix*

A BIOS Descriptions

The BIOS features addressed in this section are described as they could have a direct impact on benchmark performance. The defaults in the BIOS are denoted with a "*".

A.1 Prefetchers

There are a variety of prefetcher options available that can impact workloads. This section includes a discussion about the core prefetchers and cache prefetchers.

A.1.1 Core Prefetchers

There are several prefetchers that assist in core performance. By default, most of these are enabled, but for some workloads, it may help to disable one or more of these. Be careful changing these settings – even though some workloads can benefit from changes, the prefetchers should be viewed as a total package of settings that are chosen for the majority of day-to-day workloads.

DCU IP Prefetcher: The DCU Instruction Pointer (IP) prefetcher is an L1 cache prefetcher. Recommended default setting is Enabled. In some cases, setting this option to disabled can improve performance. Values for this BIOS option can be:

- **Disabled:** Disables the DCU IP Prefetcher
- ***Enabled:** Enables the DCU IP Prefetcher

DCU Streamer Prefetcher (a.k.a. DCU Data Prefetcher): An L1 data cache prefetcher. Recommended default setting is Enabled. In some cases, setting this option to disabled can improve performance. Values for this BIOS option can be:

- **Disabled:** Disables the DCU Streamer Prefetcher
- ***Enabled:** Enables the DCU Streamer Prefetcher

Adjacent Cache Prefetch: (a.k.a. MLC Spatial Prefetcher) is an L2 cache prefetcher. Recommended default setting is Enabled. In some cases, setting this option to disabled can improve performance. Values for this BIOS option can be:

- **Disabled:** Disables the Adjacent Cache Prefetcher
 - ***Enabled:** Enables the Adjacent Cache Prefetcher
-

Hardware Prefetcher: (a.k.a. MLC Streamer Prefetcher) is an L2 cache prefetcher. Recommended default setting is Enabled. In some cases, setting this option to disabled can improve performance. Values for this BIOS option can be:

- **Disabled:** Disables the Hardware Prefetcher
- ***Enabled:** Enables the Hardware Prefetcher

A.1.2 Last Level Cache (LLC) Prefetch

The last level cache (LLC) prefetch is a prefetcher added to the Intel Xeon Scalable processor family because of the non-inclusive cache architecture. The LLC prefetcher is an additional prefetch mechanism on top of the existing prefetchers that prefetch data into the core DCU and MLC. Enabling LLC prefetch gives the core prefetcher the ability to prefetch data directly into the LLC without necessarily filling into the L1 and L2 cache. In some cases, setting this option to disabled can improve performance. Values for this BIOS option can be:

- ***Disabled:** Disables the LLC prefetcher. The other core prefetchers are unaffected.
- **Enabled:** Gives the core prefetcher the ability to prefetch data directly to the LLC.

NOTE: The default setting for this BIOS knob may have changed compared to prior generation Intel Xeon Scalable processors.

A.1.3 Homeless Prefetch

The Homeless Prefetch allows early fetch of the demand miss into the MLC when we don't have enough resources to track this demand in the L1 cache. Special DCU prefetch which has no allocated fill buffer (FB) entry – the data is not returned to the DCU. Allows for DCU prefetching when the FB is full.

For CPUs designed with a monolithic die, Homeless Prefetch is good in most cases, as this will allow fill into the L2 cache and reduce the latency for future demand that will not pay the high latency penalty. But in some workloads these prefetch requests can overwhelm the uncore (which doesn't throttle these DCU prefetches) whereas the legacy L2 prefetches are tuned with the Dynamic feature throttling logic.

- ***Auto:** this setting maps to the hardware default setting: Disabled for XCC die, Enabled for MCC die.
- **Disabled:** Disables Homeless Prefetch, preventing early fetch from taking place.
- **Enabled:** Enables the Homeless Prefetch

A.1.4 AMP Prefetch

Adaptive Multipath Probability (AMP) prefetcher was introduced for the Golden Cove core architecture. It tracks sequences of differences between consecutive

cache accesses (delta patterns) in each local region. Using a global learning mechanism, it predicts the next delta pattern given the history of previous delta patterns and prefetches the corresponding cache lines.

The AMP prefetcher operates alongside the pre-existing L2 prefetchers: the L2 streamer and the L2 spatial prefetchers. Like these other prefetchers, AMP prefetcher brings lines to the MLC. It does not, however, issue LLC prefetches.

- **Disabled:** Disables the AMP Prefetcher
- ***Enabled:** Enables the AMP Prefetcher

A.1.5 Speculative Lock

The Speculative Lock feature is enabled by default. However, disabling Speculative Lock can yield up to ~8% improvement for some high lock-contention workloads, like SAP HANA and VMWare ESXi. Note that this benefit is not guaranteed for all high lock-contention scenarios.

Intel performs full validation with Speculative Lock enabled and recommends keeping the default (enabled) unless the above applies.

A.2 Uncore Frequency Ratio

Uncore Frequency Scaling (UFS, sometimes referred to as CLM frequency scaling), controls the frequency outside the core and is used by the mesh, LLC, memory controllers, and the mesh interface side of the IO agents and accelerators. Starting with 6th Gen Intel Xeon Scalable processor family, separate controls are provided for different types of disaggregated dies (Compute and IO).

Memory bound workloads, IO workloads or any workload with lots of accelerator device usage, as well as any workload with lots of data sharing between cores would be sensitive to uncore frequency as that will dictate the speed of data transfers between cores and memory or cache, IO and memory or cache/cores and accelerators and memory/cache/core. Scaling the uncore frequency can offer energy efficiency at the low end, while sustaining the uncore frequency at the highest frequency takes an amount of power relative to the number of supported cores.

- ***0:** Dynamically change uncore frequency between max and min fused values
- **X > 0:** Uncore will run at a constant frequency ratio of X, the UFS algorithm will be disabled, but physical limits may still reduce frequency.

A.3 Priority Core Turbo (PCT)

These knobs allow you to enable PCT, configure how many partitions of HP cores you want, and select which cores within those partitions should be prioritized. This is especially useful for AI headnode and IO-centric workloads

that benefit from maximizing single-thread performance on select cores, while other cores operate at lower frequencies to stay within the power budget. Here is a [link to PCT technical paper](#) for more information.

1. PCT Enable

Default Setting: **Enable**

Description: This knob enables or disables the Priority Core Turbo feature. When enabled, select cores (High Priority, HP) can run at elevated turbo frequencies, maximizing single-thread performance for critical workloads. If the processor does not support PCT (as indicated by a hardware capability bit), this knob and related options will be hidden.

2. PCT Partition Count

Default Setting: **4 Partitions**

Description: This knob determines how many partitions the available cores are divided into for PCT. Each partition will have designated HP cores. The default is 4, but it can be set manually. The number of partitions should not exceed the number of available PCT HP cores.

3. PCT Core Selection

Default Setting: **0**

Description: This knob specifies which core within each partition is selected as the starting HP core. By default, selection starts from core 0 (the first core in each partition). This allows fine-tuning which cores are prioritized for turbo frequency.

A.4 Snoop Mode

The Intel® Xeon® Scalable processors supports Home Snoop mode only. Home Snoop is used to launch snoops from the home agent and uses the directory (if enabled) to prevent unnecessary snoops. Home Snoop mode has two variants, without directory and with directory. Home Snoop with directory uses Opportunistic Snoop Broadcast (OSB). Home Snoop without directory is suitable for 2S configurations only.

Note: Intel recommends that customers evaluate their respective workloads on all snoop mode configurations for optimal performance.

A.5 Directory Mode Enable

The in-memory directory keeps track of the coherency of data between sockets. If the directory mode is enabled, the directory state is checked after each memory access and snoops are sent to the other socket if needed. There is only one snoop mode supported on Intel Xeon Scalable processors, home snoop, which means that the home agent is responsible for launching snoops. When directory mode is enabled, there are additional features that are used to reduce the overhead of directory reads:

OSB (opportunistic snoop broadcast) – opportunistically launches snoops before checking the directory if UPI bandwidth is available. This can reduce the latency of remote socket transfers. See the next section to learn more about using OSB on a 4-socket platform.

HitME cache – a cache of the directory state in the CHA for frequently-used lines that eliminates the need to lookup directory before launching snoops.

IODC – a cache of the directory state for IO writes and non-cacheable core writes located in the CHA. IODC allocation saves the write operation from having to do a directory lookup and update to obtain ownership of the cache line.

If directory is disabled, all memory accesses will need to snoop to check if the other socket has the line, which can cause additional overhead and is not recommended for most workloads.

Directory mode is not used in 1 socket configurations.

Values for this BIOS option can be:

- ***Auto:** same as Enable
- **Enable:** Directory is used with OSB and HitME cache.
- **Disable:** Snoops without using the directory.

Opportunistic Snoop Broadcast (OSB)

When performing a store, the directory state needs to be read from memory. The Opportunistic Snoop Broadcast (OSB) feature attempts to avoid memory read latency by snooping the local (home) agent and remote socket peers. On 2-socket systems there are multiple Intel® UPI links between the two sockets (up to 4 UPI links for SRF-SP/GNR-SP and up to 6 UPI links for SRF-AP/GNR-AP processors), therefore the link bandwidth of snoops is typically not a concern, whereas memory access could become a bottleneck.

OSB enablement is recommended on 2-socket systems for most workloads. In the reference BIOS, the Auto setting for local and remote OSB feature defaults to ON.

4- and 8-socket systems only have one Intel UPI link used between each socket, so link bandwidth of a single UPI link may be constrained between each socket pair. A large number of snoops may create a bottleneck depending on the amount of remote traffic for a given workload. Given two to four times less

Intel UPI bandwidth socket-to-socket versus 2-socket systems, the additional snoops created by the OSB feature are counterproductive. Message traffic on the local socket's internal mesh may back up as well. This scenario may cause a performance issue for several workloads including:

- in-memory databases with substantial remote traffic
- streaming store workloads.
- ***Auto:** same as Enable
- **Enable:** Directory is used with OSB and HitME cache
- **Disable:** Snoops without using the directory

In the reference BIOS, Auto may not distinguish between 2-, 4- and 8-socket platforms. Intel recommends that the BIOS is modified to keep Auto as the default and then make the following decisions if Auto is selected:

1. Determine the # of sockets
2. For 2-socket, <**Enable**/Disable> OSB
3. For 4- and 8-socket, <Enable/**Disable**> OSB

NOTES: This feature is usually not exposed in customer UEFIs. Please work with your local Intel BIOS expert if further details are needed.

Directory A-to-S (Stale AtoS or A2S)

Stale AtoS (sometimes called A2S) is the transition of a directory line state. The in-memory directory has three states: I, A, and S. Invalid (I) state means the data is clean and does not exist in any other socket's cache. snoopAll (A) state means the data may exist in another socket in exclusive or modified state. Shared (S) state means the data is clean and may be shared across one or more socket's caches. When doing a read to memory, if the directory line is in the A state we must snoop all the other sockets because another socket may have the line in modified state. If this is the case, the snoop will return the modified data. However, it may be the case that a line is read in A state and all the snoops come back a miss. This can happen if another socket read the line earlier and then silently dropped it from its cache without modifying it. Values for this BIOS option can be:

- ***Auto:** Recommended setting (same as Enable)
- **Disable:** Disabling this option allows the feature to process memory directory states as described previously.
- **Enable:** In the situation where a line in A state returns only snoop misses, the line will transition to S state. That way, subsequent reads to the line will encounter it in S state and not have to snoop, saving latency and snoop bandwidth.

Stale A-to-S may be beneficial in a workload where there are many cross-socket reads. For greater than two-socket configurations, it is recommended that this knob be enabled. This can result in improved remote latency and bandwidth in certain situations, particularly when the Intel® UPI link is heavily utilized.

IO Directory Cache (IODC)

IODC allows the remote socket to suppress directory read/update for non-cacheable write transactions. It generates snoops instead of memory lookups, for remote InvItom (IIO) and/or WCiLF (cores). When enabled, this reduces directory-based write overhead (e.g., high memory bandwidth demanding non-NUMA micros benefit 10-90%). On prior Intel Xeon processor generations, this has been enabled for IO writes. Values for this BIOS option can be:

- ***Auto:** same as Enable
- **Disable:** don't suppress directory read/updates for non-cacheable write transactions
- **Enable for Remote InvItom Hybrid Push**
- **Enable for Remote InvItom AllocFlow**
- **Enable for Remote InvItom Hybrid AllocNonAlloc**
- **Enable for Remote InvItom and Remote WCiLF**

A.6 On-die Clustering Options

6th Gen Xeon processors support multiple modes for clustering the Last Level Cache (LLC) slices on the socket in non-uniform memory access (NUMA) interleave mode. These modes divide up the LLC slices on the die into one of 'n' different cluster patterns. The difference between the modes is in how the cluster for a given transaction is chosen. One method divides the die into sub-NUMA domains and uses address range checks to determine the cluster. This mode is called Sub-NUMA Clustering (SNC) and was first introduced in the first generation of the Xeon Scalable processor family. Beginning with the 3rd Gen Xeon Scalable family, SNC2 was introduced.

- SNC3 and SNC2 are available on select Granite Rapids processors.
- SNC is not available for Sierra Forest-SP processors.

The other two methods use lower address bits to select the cluster and ends up interleaving the address space across the clusters within a die in a UMA fashion. These two UMA modes, called Hemisphere and Quadrant, provide latency advantages without software applications needing to be NUMA-aware.

Values for the NUMA BIOS option can be:

- ***Auto:** default set to match SNC setting
- **Disable:** default is UMA mode when SNCx is disabled.
- **Enable:** NUMA interleave mode when SNCx is enabled.

For virtualization configurations (e.g., VMware*), enabling SNC could improve performance, but it depends on the size of the VM. If the VM is larger than any die tile and the direct memory associated with that tile, SNC should be disabled. But if the VM is less than a tile's cores and memory then SNC is needed, and the default UMA setting should not be used.

A.6.1 2-way Sub-NUMA Clustering (SNC2)

Sub-NUMA Clustering (SNC or SNC2) is only supported on GNR-SP XCC/HCC and partitions the LLC into two clusters called NUMA domains (a.k.a. nodes) that contain an equal number of cores, equal number of LLC slices in close proximity to the cores, an equal amount of socket address space, and with each cluster bound to a subset of the memory controllers in the socket. SNC mode requires software support. To make effective use of SNC mode the operating system must view each SNC range as a separate NUMA range (hence the name Sub-NUMA cluster) and be sure to schedule the threads to the cores nearest the cluster to which its memory is allocated. In this way, SNC mode partitions not only the LLC slices, but also (at the software level) the cores.

Each SNC is an independent NUMA node visible to software. With sub-NUMA clustering, the way the LLC request is routed is determined by which cluster contains the agent for the address. This means that if a core in cluster 0 issues a read to an address in cluster 1, the request is serviced by cluster 1's LLC. In this sense, with SNC, the cache is still technically a single unified cache since every system address maps to one and only one of the two cluster's LLC. Access was considered remote for COD for some non-NUMA accesses, but for SNC, the data can be placed in the MLC. To state this another way, addresses from the remote cluster never get cached in the local cluster LLC. In some cases, this can result in a slightly larger latency compared to the COD latency.

Each NUMA node is an independent NUMA node visible to the OS or the software application.

- **Auto:** Same as Enabled
- **On:** SNC On. Enables Sub-NUMA Clustering of SNC2
- **Off:** SNC off. Defaults to HEX mode (see A.5.3 on how to change to All2All)

NOTE: See the technical paper *Granite Rapids AP Clustering Modes*, for a complete description of SNC for GNR-AP/SP. <doc# goes here>

A.6.2 3-way Sub-NUMA Clustering (SNC3)

SNC3 is only supported on GNR-AP and partitions the LLC into three clusters. This option is available for Granite Rapids UCC processors, which have 3 compute dies.

- **Auto:** Same as Enabled
- **On:** SNC On. Enables Sub-NUMA Clustering of SNC3
- **Off:** SNC off. Defaults to HEX mode (see A.5.3 on how to change to All2All)

NOTE: See the white paper *Granite Rapids AP Clustering Modes*, for a complete description of SNC for GNR-AP/SP. <doc# goes here>

A.6.3 VirtualNUMA Configuration

More details in [whitepaper](#) (RDC Doc ID 845720)

Divides physical NUMA nodes into evenly sized virtual NUMA nodes in ACPI table. Enabling this feature improve Windows server application performances on CPUs with more than 64 logical processors per numa.

VirtualNUMA provides same local latency for each VirtualNUMA node and max interleaving bandwidth for a sub-NUMA configuration that is set on a platform. Unlike SNC, VirtualNUMA doesn't break up the last level cache (LLC) into disjoint clusters. In a sense, with VirtualNUMA, the LLC cache routing is still technically uniform with respect to any sub-numa configuration settings that is set on a platform.

Enabling this feature provides the option to set number of VirtualNuma nodes per physical NUMA nodes

- 0 means automatically set the number of virtual NUMA nodes based on system configuration.
- N - User configurable to N VirtualNuma nodes.

SNC	VirtualNuma	Notes
*Disable	Enable	• SRAT reports Multiple Proximity domains (Vnuma nodes) with less than 64LPs. ○ Each Vnuma owns equal memory capacity. ○ Inherits quad or hex channel interleaving (if all channels populated) ○ LLC access routing uniform via all memory controller within socket.

Snc2 Enable	Enable	- SRAT reports Multiple Proximity domains (Vnuma nodes) with less than 64LPs per numa on top of SNC2 physical numa nodes - Socket= SNC2 numa nodes + Vnuma nodes - Each Vnuma owns equal memory capacity. - Inherits SNC2 channel interleaving (if all channels populated) - Inherits SNC2 LLC access routing via memory controller split.
SNC3 Enable	Enable	- SRAT reports Multiple Proximity domains (Vnuma nodes) with less than 64LPs per numa on top of SNC3 physical numa nodes - Socket= SNC3 NUMA nodes + Vnuma nodes - Each Vnuma owns equal memory. - Inherits SNC3 channel interleaving (if all channels populated) - Inherits SNC3 LLC access routing via memory controller split.

For Windows SQL server database application with Granite Rapids and Sierra Forest processor highly recommend enabling VirtualNuma for the below scenarios.

GNR-SP	Intel recommended bios options for windows SQL server*
Less than 64core SKU	Snc2
Greater than 64core SKU	Snc2+VirtualNuma

GNR-AP	Intel recommended bios options for windows SQL server*
Less than 96core SKU	Snc3
Greater than 96core SKU	Snc3 + VirtualNuma

SRF-SP	Intel recommended bios options for windows SQL server*
144core	Hex + VirtualNuma

Windows SQL server - Even though Windows server 2022 and above has no limitation with number of logical processor per numa. Microsoft SQL Server (2012-2022*) still limits the number of logical processors per NUMA node to 64. Current on-die options snc2 or snc3 clustering on certain Xeon6 high core count SKUs is not efficient solution for application.*

A.7 TOR Thresholds (Loc2Rem)

In multi-socket environments that generate remote traffic, the Table Of Requests (TOR) in the CHA services both remote and local requests across the Intel® UPI. Remote reads allocate the TOR through the RRQ ingress and

typically send out a read request to the iMC. When the data returns from the iMC, the TOR will send a data response back to the requesting socket via the Intel® UPI. The rate at which these requests and responses can be sent out is governed by Intel® UPI bandwidth via credits. The rate is adjustable to limit the number of incoming transactions to avoid overloading a remote socket. This bandwidth is substantially less than local memory bandwidth or TOR service bandwidth. When Intel® UPI is the bottleneck, these remote reads stay in the TOR for a long time. When one socket's TOR is saturated with remote reads, it cannot service its own local requests well enough. The other socket is relatively free to continue sending remote requests to the saturated socket, which may cause an imbalance. If the imbalance shifts around then the performance averages out over time and can converge to a balance, but if a socket is saturated, and continues to get requests from the better performing socket, it may not build enough momentum to overturn the imbalance in its favor. Application workloads with UMA and remote traffic mixes are likely candidates for this imbalance and could affect overall performance.

The BIOS option provides a set of thresholds that can control how much of the various types of requests are allowed to occupy the TOR, thus helping to avoid the imbalance. This BIOS option controls the number of local-to-remote requests allowed in the pipeline when the pipeline is empty of remote requests (EMPTY) and when remote requests are also present in the pipeline (NORMAL).

- ***Auto:** see Recommended TOR *Threshold Settings for Multi-Socket Platforms* based on the number of platform sockets populated.
- **Disable:** Loc2Rem values are disabled
- **Low, Medium, High:** See table based on the number of platform sockets populated.

These features are used to control bandwidth of a multi-socket system, to maximize the throughput of each single socket while trying to make the performance across sockets as balanced as possible. The default threshold values set by BIOS are defined based on internal experiments and are believed to be optimal for typical workloads.

The threshold values set by the BIOS are dependent on the system configuration. The BIOS will first detect the platform configuration in terms of the number of sockets and the number of Intel® UPI links and then auto set default mode accordingly. Recommended settings are shown below for low, medium, and high bandwidth workloads.

Table Error! No text of specified style in document.-1. Recommended TOR Threshold Settings for Multi-Socket Platforms

Number of Sockets	Auto (default)		Disable	Low	Medium	High
1S, 2S						
4S						
8S						

--	--	--	--	--	--	--

NOTES: Intel recommends that customers evaluate their respective workloads, when considering these threshold configurations for optimal performance. These are listed as recommended settings, and they are subject to change.

A.8 XPT Prefetch

Local XPT Prefetch is a feature designed to reduce local memory access latency. When enabled, upon receiving a read request from the core, the core-uncore interface sends a speculative read request directly to the memory controller in parallel with the read request to the LLC, assuming the LLC lookup will miss. If the LLC lookup is a hit, the XPT prefetch data is discarded by the memory controller. Bypassing the LLC lookup ultimately leads to a reduction in latency.

XPT and UPI provide similar latency reductions, but only one can be effective at a time. If the UPI Prefetcher is enabled, the XPT remote prefetcher's benefit will not be visible. Disabling the UPI Prefetcher allows the XPT to reduce latency. Additionally, for the XPT remote to be functional, the XPT local must also be enabled.

By default, the UPI prefetcher is enabled, making the XPT remote ineffective. The BIOS configuration is optimized for remote latency cases.

- ***Auto:** same as Enabled
- **Disabled:** Sets the disable bit of XPT_LOCAL_PREFETCH_CONFIG1 (bit0)
- **Enabled:** Clears the disable bit of XPT_LOCAL_PREFETCH_CONFIG1 (bit0)
- **If XPT Prefetch = Enabled, then remote prefetch can be either value.**
- **But if XPT Pretetch = Disabled, then remote prefetch being enabled does not provide any benefits.**

A.9 Dead Line LLC Allocation

In the Intel® Xeon® Scalable family non-inclusive cache scheme, MLC evictions are filled into the LLC. When lines are evicted from the MLC, the core can flag them as "dead" (in other words, not likely to be read again). The LLC has the option to drop dead lines and not fill them in the LLC. This can help save space in the LLC and prevent the LLC from evicting useful data. Values for this BIOS option can be:

- **Disabled:** Disabling this option can save space in the LLC by never moving MLC dead lines into the LLC.
- ***Enabled:** Opportunistically move MLC dead lines into the LLC, if space is available.
- **Auto:** BIOS will choose the correct setting

A.10 MSI Bypass

Intel Xeon 6 processors provide a configurable option (MSI Bypass) to allow subsequent DMA writes and reads to be processed without waiting for earlier MSIs to be delivered to cores. Enabling this mode does not affect the handling of completions, which are never allowed to bypass MSIs.

The implementation of the MSI Bypass feature requires the activation of both a global knob and per-PCIe stack knobs in the BIOS. Specifically, the following configuration steps are needed:

1. **Global Configuration:** The global MSI Bypass knob must be enabled through the IIO Global Configuration menu by toggling the 'Toggle MSI Bypass' setting.

EDKII menu-> Socket Configuration-> IIO Global Configuration-> IIO Global Configuration -> (Under Misc Global) Toggle MSI Bypass

2. **Per-Stack Configuration:** For each individual PCIe stack the corresponding MSI Bypass knob must be enabled through the Socket0/1 Configuration -> PCI Express [Number] -> 'MSI Bypass' Enable settings.

EDKII menu-> Socket Configuration-> IIO Global Configuration-> Socket[Number] Configuration->PCI Express [Number] -> MSI Bypass.

Both the global Toggle MSI Bypass knob and the respective per-stack knobs need to be enabled to fully activate the MSI Bypass feature.

If the MSI bypass feature is enabled, additional safeguards are required for software processes that rely on draining of interrupts. For example, prior to moving a guest VM to a new core, software must ensure that interrupts sent using the prior values in the Interrupt Remapping Table Entry (IRTE) have drained, i.e., become observable.

For more details please refer to "17.1.5 MSI Bypass Performance Optimization" in the Intel Xeon 6 processors technical digest [#850472](#).

A.11 Clock Modulation

This knob is used to read and write the duty cycle bits, and to enable the on-demand clock modulation feature of the Intel Thermal Monitor. Values for this BIOS option (Socket Configuration > Uncore Configuration > Uncore General Configuration > Clock Modulation Enabled) can be:

- ***Auto:** Recommended setting (same as Enable).
 - **Disable:** When clear, disables the software-controlled clock modulation.
-

- **Enable:** When set, enables on-demand software-controlled clock modulation.

A.12 Intel® Ultra Path Interconnect

For SP platforms there are up to four links per processor and for AP platforms, up to six links per processor are available. Each link is 24bits wide. The UPI speed is either 16, 20, or 24GT/s.

A.12.1 Link Speed Mode

This knob allows for selecting the Intel UPI 2.0 link frequency. Options are:

- ***Auto:** Auto decides based on Si Compatibility. Recommended setting.
- **Slow:** Keep slow speed
- **Fast:** Train the UPI link to Fast speed

A.12.2 UPI Prefetch

Intel® Ultra Path Interconnect (Intel® UPI) Prefetch (a.k.a. KTI Prefetch) is another capability that is designed to reduce remote memory access latency. The Intel UPI controller issues a UPI prefetch to the memory controller when a remote read arrives to the home socket. The Intel UPI controller issues the Intel UPI prefetch in parallel to an LLC lookup, similar to the XPT prefetch. Data from a Intel UPI prefetch will wait for a short time in the memory controller and will be returned to the Intel UPI in the event of an actual memory read. As with XPT prefetch, bypassing the LLC lookup results in a reduction in latency. NOTE: UPI prefetch works in a similar fashion as Local XPT prefetch, except that it applies to reads coming from the UPI instead of the core.

- ***Auto:** BIOS decides correct setting
- **Disable:** Disables KTI Prefetch
- **Enable:** Enables KTI Prefetch

A.12.3 Direct-to-UPI (D2U)

D2U (or D2K) is a latency-saving feature for remote read transactions. After the iMC processes a read coming from a remote socket, it will send the data back to the Caching and Home Agent (CHA), which will then send it to the Intel® UPI so the data can go back to the requester socket. When the D2U feature is used, the iMC will send the data directly to the Intel® UPI instead of going through the CHA. This reduces latency.

D2U uses a crediting scheme between the memory and the Intel® UPI ingress. There is a fixed number of Intel® UPI ingress credits allocated for D2U transactions. So, there is a fixed maximum number of D2U transactions that can be outstanding at any given time.

- ***Auto:** <what happens when auto is selected>
-

- **Disabled:** Disabling D2U causes the iMC to send data via its local CHA to the Intel® UPI.
- **Enabled:** When D2U is enabled, and during remote read transactions, the iMC sends data directly to the Intel® UPI, which results in latency savings.

Workloads that are highly dependent on remote idle latency may see an impact when D2U is disabled. Workloads that are highly NUMA-optimized or that use high levels of memory bandwidth are less likely to be affected by disabling D2U.

A.13 Integrated I/O (IIO) Settings

Tuning from the BIOS can be done to improve I/O performance and power efficiency in most cases using the following BIOS settings.

Advance Menu BIOS Settings

Menu (Advanced)	Path to BIOS Settings		BIOS Setting	Recommend Settings for Performance
Power Configuration	Socket Configuration > Advanced Power Management Configuration > CPU P State Control	Enhanced Intel SpeedStep® Technology PSD Function		HW_ALL
			Intel® Turbo Boost Technology	Enable
			Intel SpeedStep® (P-States)	Enable
	Socket Configuration > Advanced Power Management Configuration > Hardware PM State Control	Hardware P-States		Native Mode
	Socket Configuration > Advanced Power Management Configuration > CPU C State Control	Enhanced Halt State (C1E)	Enable	
	Socket Configuration > Advanced Power Management Configuration > Package C State Control		Package C State	C0/C1/C6 State
	Socket Configuration > IIO Configuration > Global Configuration		Relax Ordering	Disable

Menu (Advanced)	Path to BIOS Settings	BIOS Setting	Recommend Settings for Performance
I/O Configuration	Socket Configuration > I/O Configuration > Intel VT for Directed I/O (VT-d)	Intel® VT-d	Enable
Intel® UPI Configuration	Socket Configuration > Uncore Configuration > Uncore General Configuration	LinkINK L1 Enable	Disable

A.13.1 Inbound Flows

Inbound Write Allocating – Default Setting Per Root Port

The allocating flow is an inbound write flow that is meant to be allocated in the LLC. This is useful in workloads that have core and I/O cooperation, for example, cores consuming I/O produced data with minimum latency. By having the data in the LLC, the core access time to I/O produced data is reduced.

For the full line write, an I2M request is issued from I/O to the CHA when there is an inbound IIO write that is going to write an entire cache line. The CHA gets ownership of the line and delivers ownership to the IIO. There is no data delivery to I/O in this phase, but the ownership request may cause write-back eviction to DRAM if data is in M state in LLC.

Once ownership is received, the inbound traffic controller which maintains ordering for all inbound writes will complete the write to the I/O write cache. Once it is in the write cache, it is globally observable and snoopable by other coherency agents.

The ownership phase of the full line write could be out of order, but the IIO block enforces a PCIe* order during the write-back phase, when the data is committed to the I/O write cache.

Example:

If we get 10 writes coming up to the PCIe* interface, all in strict order (RO attribute on PCIe* is 0) – the first write blocks the second write, then this blocks the third write and so on, down to the ordering line.

All 10 writes may issue I2M ownership requests nearly simultaneously from write cache. Responses to I2M – Go-E – will come back out of order as well. There is no guarantee that Go-Es will come back in the PCIe* order from all the different CHAs ownership request it may target.

The head of the ordering queue is the first write request and that first write is allowed to commit its data via the write-back phase as soon it receives its ownership via GO-E message. It then gets tagged as “Modified” and can be issued to coherency domain by the write cache.

The second write in the order queue must wait for the first write to finish its Request for Ownership (RFO) phase and is posted to write cache before it can issue its own write-back request.

Due to the unordered nature of the coherency domain, the GO-E message indicating the ownership phase has completed will be out of order with respect to the PCIe* order. In that case, the writes cannot commit until the request at the head of order queue is committed.

If there is a write that hits the head of the queue and its Go-E does not come back in a timely manner, the head of queue will stall and will result in the Head Of Line (HOL) blocking and the BW will be lost.

For a partial line, the IIO is responsible for combining a small amount of data that gets on the PCIe* interface with full cache line. To do that, it issues an RFO request, the response to an RFO from the CHA is Go-M or Go-E and there is data delivered back to the IIO.

When the partial write reaches the head of the ITC queue, the IIO will merge that data in the write cache to form a complete cache data line and then the write-back phase will write the complete cache line back to the system.

In case of a full line write or a partial line write, when the IIO writes back the data to the CHA, the CHA deposits the data in the LLC. This feature is what we call Data Direct I/O because we deposit data that is I/O generated traffic into the LLC, and it is immediately accessible to cores that may want to access that data within a reasonable amount of time.

There is careful management from the software between allowing this to happen for certain I/O traffic and sending another flow (non-allocating) to prevent the I/O data from polluting the cache. The other thing that helps to limit the LLC pollution is that the I/O traffic is only allowed to write into two ways of the LLC by default. That is also configurable via the MSR for IIO_LLC_WAYS.

For usages that could benefit from more IO ways in the LLC than the default, this MSR can be written to increase the I/O ways.

The I/O LLC ways can be written into by cores, whereas I/O cannot write into core ways. Also, if the I/O's PCIe* write results in a LLC hit because the address has previously been used by core, the I/O write can use that LLC location, even if it is not an I/O way.

IIO LLC Ways

19.4.580 (C8Bh) IIO_LLC_WAYS

This register controls the LLC way mask for Intel Data Direct I/O Technology (DDIO)

Type	Size	Scope	Offset	Default
MSR	64 bit	package	C8Bh	000000000000FFFFFh

Register Level Access:

Read Access	Write Access
R	W

Bit Range	Default and Access	Field Name (ID): Description
63:32	0h	Reserved

Bit Range	Default and Access	Field Name (ID): Description
	RO	
31:0	000FFFFh RW	WAY_MASK: Available LLC ways vectors for class of service of IIO

Inbound Write Non-allocating

This is what we use for delivering data to the DRAM. It means that we are not going to allocate the LLC entry for the data.

When the write-back from the I/O is sent to the CHA, the CHA is just going to evict it to the DRAM and write to the DRAM immediately. The flow begins the same way as the allocating the Intel® Data Direct I/O Technology (Intel® DDIO) flow with the I2M or the RFO for the ownership phase. In the write-back phase, I/O issues the CLFlush instead of the WBM2I and data is flushed from the LLC to the DRAM.

The in-bound allocating or non-allocating mode is set via the IIMISCCTRL CSR for the entire I/O stack or controlled via the PERFCTRLSTS CSR per individual Root Port.

In addition, customers can have per transaction level (Translation Layer Packet [TLP] level granularity from PCIe*) by setting up the MISCCTRL or the PERFCTRLSTS for allocating but setting the NS bit in the PCIe* TLP header to one for the transactions that need to be flushed from the cache. When the IIO stack detects that NS=1 setting, it will flush that transaction from the LLC to the DRAM.

Inbound Read

All inbound reads will use the RdCurr IDI opcode. This will read the cache line wherever it sits and not change its coherency state.

It does have one side effect, if the read line is in the L2 cache, it will cause eviction out of the L2 cache to the LLC. Another thing to note is if it is pulled from the DRAM then it stays in the DRAM and not in the LLC.

Enabling Allocating and Non-allocating Flow

The `disable_all_allocating_flows` setting trumps all other settings; this setting is per I/O stack. There is one IIMISCCTRL CSR per I/O stack.

IIO MISC Control

IIMISCCTRL: IIO MISC Control			
Bus: RootBus0		Device: 5	Function: 0
		Offset: 1C0h	
Bit	Attr	Default	Description
24	RW	0b	Disable_all_allocating_flows When this bit is set, IIO will no more issue any new inbound IDI command that can allocate into LLC. Instead, all the writes will use one of the non-allocating commands - PCIWIL/PCIWILF/PCINSWr/PCINSWrF. This is provided primarily for PSMI where we need a mode to not allocate into the LLC. Software should set this bit only when no requests are being actively issued on IDI. So either a lock/quiesce flow should be employed before this bit is set/cleared or it should be set up before DMA is enabled in system.

PERFCTRLSTS CSR: there is one CSR per RP. In this CSR, there is a use_allocating_flow_wr bit. When it is set to one, allocating write is enabled.

In addition, the NoSnoopOpWrEn allows a PCIe* device attached to the RP to set the NS attribute to one and the IIO will honor that attribute through the non-allocating flow.

PERFCTRLSTS: Performance Control and Status, Bit 3

PERFCTRLSTS: Performance Control and Status			
Bus: RootBus0		Device: 0	Function: 0
		Offset: 180h	Mode: DMI
Bit	Attr	Default	Description
3	RW	0b	NoSnoopOpWrEn Enable No-Snoop Optimization on VC0 writes and VCp writes This applies to writes with the following conditions: NS=1 AND (TPH=0 OR TPHDIS=1) 1: Inbound writes to memory with above conditions will be treated as non-coherent (no snoops) writes on Intel® Ultra Path Interconnect (Intel® UPI) 0: Inbound writes to memory with above conditions will be treated as allocating or non-allocating writes, depending on bit 4 in this register. Notes: - If TPH=1 and TPHDIS=0 then NS is ignored and this bit is ignored - VC1/VCm writes are not controlled by this bit since they are always non-snoop and can be no other way. - Current recommendation for BIOS is to just leave this bit at default of 0b.

When nosnoopopwren is set to zero, the NS PCI Header bit is a “don’t care.”

Performance Control and Status, Bit 7

PERFCTRLSTS: Performance Control and Status			
Bus: RootBus0		Device: 0	Function: 0
		Offset: 180h	Mode: DMI
Bit	Attr	Default	Description
7	RW	1b	UseAllocatingFlowWr Use Allocating Flows for 'Normal Writes' on VC0 and VCp 1: Use allocating flows for the writes that meet the following criteria. 0: Use non-allocating flows for writes that meet the following criteria (TPH=0 OR TPHDIS=1 OR (TPH=1 AND Tag=0 AND CIPCTRL[28]=1)) AND (NS=0 OR NoSnoopOpWrEn=0) AND Non-DCA Write Notes: - Allocating and non-allocating flows are described in Transaction Flow Chapter - See Transaction Flow Chapter for writes that qualify as Non-DCA writes. - VC1/VCm traffic is not impacted by this bit in Dev#0 - When allocating flows are used for the above write types, IIO does not send a Prefetch Hint message. - Current recommendation for BIOS is to just leave this bit at default of 1b for all but DMI port. For DMI port when operating in DMI mode, this bit must be left at default value and when operating in PCIe mode, this bit should be set by BIOS. - Note there is a coupling between the usage of this bit and bits 2 and 3. - TPHDIS is bit 41 of this register - NoSnoopOpWrEn is bit 3 of this register

These options can be condensed into the following state table:

State Table

Option	disable_all_allocating_flows Bit 24 (PER IIO Stack)	Nosnoop opwren Bit 3 (PER ROOT-PORT)	Use_allocating_flow_wr Bit 7 (PER ROOT-PORT)	PCI Transaction Heder NS Bit (PER PCI IO)	Resulting IIO Write	Intel® DDIO	Notes
1	1	X	X	X	Non-Allocating	NO	Per stack global control to do non-allocating
2	0	0	0	X	Non-Allocating	NO	OPTIONAL PER ROOT PORT

Option	disable all allocating flows Bit 24 (PER IIO Stack)	Nosnoop opwren Bit 3 (PER ROOT- PORT)	Use allocating _flow_wr Bit 7 (PER ROOT- PORT)	PCI Transact ion Heder NS Bit (PER PCI IO)	Resulting IIO Write	Intel® DDIO	Notes
2	0	0	1	X	Allocating	YES	DEFAULT SETTING PER ROOT PORT
3	0	1	1	0	Allocating	YES	Enable Per Root and OPTIONAL PER PCI TRANSACTION
3	0	1	1	1	Non- Allocating	NO	

A.13.2 Peer to Peer (P2P)

There are two types of P2P flow in the Intel® Xeon® Scalable processor family.

The first is the confined P2P. In this case, the P2P is from a bifurcated root port to another bifurcated root port in the same I/O stack. The P2P traffic is handled entirely within the I/O stack. The performance expectation is that each of the bifurcated ports will get the PCIe* line a rate performance of 64-B packets.

The other type of flow is from one root port in a particular I/O stack to another root port in a different I/O stack. The IIO RP will break the P2P requests into 64-B packets (if the P2P requests are larger than 64-B) and send out a NcP2PB packet to the other I/O stack through the mesh. The performance expectation for the link is to achieve a PCIe* line rate performance of 64-B packets.

For P2P flows, there are some special usages that may require end user or platform tuning. One potential usage is seen with some Nvidia* Graphic Processing Unit (GPU) cards, which do a P2P involving both the MMIOH and the MMIOH regions. In these cases, the following CSRs need to be checked:

MMIOL_RULE*_5_0_CFG, bits 2:1 indicate Serialize Rule:

MMIOL_RULE*_5_0_CFG

CSR Name	Bus	Device	Function	Offset	Bits	Field Name
MMIOL_RULE	x	5	0	0x300-0x32C	2:1	Serialize Rule

MMIOH_NON_INTERLEAVE*_5_0_CFG, bits 2:1 indicate Serialize Rule:

MMIOH_NON_INTERLEAVE*_5_0_CFG

CSR Name	Bus	Device	Function	Offset	Bits	Field Name
MMIOH_NON_INTERLEAVE	x	5	0	0x340	2:1	Serialize Rule

MMIOH_INTERLEAVE_*_5_0_CFG, bits 2:1 indicate Serialize Rule:

MMIOH_INTERLEAVE_*_5_0_CFG

CSR Name	Bus	Device	Function	Offset	Bits	Field Name
MMIOH_INTERLEAVE	x	5	0	0x348	2:1	Serialize Rule

By default, the Intel® Xeon® Scalable family will maintain strict PCIe* order on traffic between the MMIOL region and the MMIOH region.

If the P2P usage switches between the two regions, then the Serialize Rule in these CSRs needs to be programmed to indicate that the MMIOL and the MMIOH are toward the same target and not to be serialized. In these cases, both the MMIOL and the MMIOH Serialize Rule fields need to be programmed to one.

In this specific usage, the MMIOH region is interleaved amongst different PCIe* or I/O stacks. Also, if the P2P flow is across multiple PCIe* or I/O stacks with the MMIOL and the MMIOH mix, then these changes do not improve performance.

A.13.3 Outbound Flows

For outbound MMIO write flows, it is recommended that the core instructions use write combining UC writes or use the Intel® Advanced Vector Extensions 512 (Intel® AVX-512) instructions to get 64-B outbound packets for maximum PCIe* outbound BW.

Snoop Case

When the I/O has ownership of a line in the LLC indicated by `CV.IIO bit =1`, when a snoop for I/O owned line happens, the CHA will forward that snoop to the IIO. When the I/O receives the snoop, it must give up ownership of the line.

However, if the PCIe* order for that line has not been achieved, then the I/O has to reacquire ownership for that line again. Performance drops because the I/O has to spend extra cycles to regain ownership and wait for the PCIe* order.

An example of this type of workload is the Data Plane Development Kit* (DPDK*) network test. In this set up, multiple network cards are transmitting and receiving data. To process packets at extremely high rate (a packet every 70 ns in the case of one 10GbE port), the core constantly checks for new incoming packets by polling the receive descriptor write-back status of the 10GbE port.

The DPDK* driver uses a thread to poll the descriptor status continuously in a tight loop so that the incoming packets can be processed as fast as possible. This creates a core and I/O contention for the I/O owned line. The underlying contention causes the I/O to lose ownership of the LLC lines it has just

acquired due to core snoop and causes the I/O to spend more cycles to reacquire the line again to write it back.

In cases where there are large of 10GbE ports each servicing small packets, the test may not be able to achieve the full throughput required on the network due to the excessive I/O and core contentions arising for each port. In these cases, it makes sense for the I/O to delay its responses to core snoops and hold on to its cache lines until it can dispatch its write successfully.

In a standard platform BIOS, there is a menu under the I/O options that shows different options to delay the I/O snoop response. Users can experiment with different values to achieve their best performance with each specific usage.

A.13.4 Intel® DDIO Usage Tips

While the Intel® DDIO gives some of the LLC ways for the I/O to deposit write data, there are some considerations that need to be taken into account to achieve maximum BW.

The first consideration is that the LLC is a cache, and usage of cache is indexed by the address and the temporal locality of the data.

In traditional I/O usages with high streaming BW, software or hardware tends to treat a region of memory as a buffer, with incrementing address. In these cases, if the buffer is constantly reusing or recycling a set of addresses, it may cause unnecessary LLC evictions and lowering the I/O BW.

Fortunately, these types of workloads are normally artificial benchmarks and not something seen in realistic usage. In realistic usage, OSes would often assign addresses and buffers that do not limit the LLC sets, and there are often core and I/O interactions such that this type of eviction does not happen.

This LLC behavior is not limited to I/O. Non-IO workloads also face the same LLC constraints. For these workloads, the SW community has not been forcing address patterns that are this artificial, and have a good history of carefully crafting realistic test patterns that test actual performance.

A.13.5 Common Buffer Direct Memory Access (CBDMA) Performance

The capabilities of CBDMA are now available by using the Intel DSA accelerator. See section C.4 Intel® Data Streaming Accelerator (Intel® DSA) for more details.

A.13.6 Non-Transparent Bridge (NTB) Performance

The NTB is a PCIe* to PCIe* link between two OS instances. Each side of the link sees one PCIe* endpoint attached to a root port, and the data is transferred between the two endpoints. The NTB hardware translates the addresses between the two end points via a set of Base Address Registers (BARs) for primary and secondary addresses.

For more details on the NTB operation, consult the NTB chapter in the Intel® Xeon® Processor Scalable Family EDS.

For the NTB to operate as a bi-directional link, the `MISCCTRLSTS_1_CSR` for each NTB stack needs to have bit 6, and `problematic_port_for_lock_flows` set to one.

A.13.7 Asynchronous Dynamic Refresh (ADR)

The ADR refers to when memory is in self-refresh mode. For systems that require data to be maintained in battery backed up DRAM or flash, the I/O flows need to be sent to the memory controller domain prior to the system being powered down or a link going down. For this type of usage, a non-allocating flow is recommended.

In the current Intel® Xeon® processor Scalable family, the non-allocating flow is improved from previous generations due to the memory controller Cmp being propagated to IIO through the `GO_I` indication at end of the CLFlush phase. For more details, see the previous non-allocating flow.

Another update to these processors is the use of NS hint in the PCIe* TLP header to invoke the non-allocating flow. This allows the end user the flexibility to push certain transactions that are pushed to memory and others that maybe allocated in LLC.

A.14 Intel® Advanced Vector Extensions and Intel® Advanced Matrix Extensions

A.14.1 Recommended Performance Optimizations for Intel® Advanced Vector Extensions 2 (Intel® AVX2)

See the white paper on how to optimize performance for Intel® AVX2 instructions on the Intel® Xeon® processor Scalable family:

[How Intel® Advanced Vector Extensions 2 Improves Performance on Server Applications.](#)

A.14.2 Recommended Performance Optimizations for Intel® Advanced Vector Extensions 512 (Intel® AVX-512)

The release of the Intel® Xeon® processor Scalable family includes the Intel® AVX-512 instruction set, which can provide significant performance improvement over Intel® AVX2, Intel® AVX, and Intel® Streaming SIMD Extensions (Intel® SSE). The Intel® AVX-512 instructions require more power to run. When executing these instructions, the processor may run at less than the marked frequency to maintain the Thermal Design Power (TDP) limits.

- Intel AVX-512 is available in Intel Xeon 6 processors with P-cores

To learn more about Intel AVX-512 using our Development Reference Guides, visit <https://www.intel.com/content/www/us/en/develop/documentation/cpp-compiler-developer-guide-and-reference/top/compiler-reference/intrinsics/intrinsics-for-avx-512-instructions.html> where there's a discussion about intrinsics, programming and compiler optimizations.

A.14.3 Recommended Performance Optimizations for Intel® Advanced Matrix Extensions (Intel® AMX)

Intel AMX is a set of matrix multiply instructions that can operate on a tile using a variety of datatypes including bfloat16 (BF16) and int8 to accelerate deep learning inference and training tasks.

To learn more about Intel AMX using our Development Reference Guides, visit (<https://www.intel.com/content/www/us/en/develop/documentation/cpp-compiler-developer-guide-and-reference/top/compiler-reference/intrinsics/intrinsics-for-amx-instructions.html?wapkw=AMX%20advanced%20matrix%20extensions>) where there's a discussion about intrinsics, programming and compiler optimizations.

For performance- and productivity-optimized tools from Intel that can help you develop, train, and deploy your AI solutions quickly, visit [AI Tools, Libraries, and Framework Optimizations](#). Tools are provided for data preparation, training, inference, deployment, and scaling and are built on the foundation of a standards-based, unified oneAPI programming model with interoperability, openness, and extensibility.

- Intel AMX is available in p-core processors
-

Enable LP [Global] (Intel® Hyper-Threading Technology)

The Intel Hyper-Threading knob has been renamed Enable LP [Global] to represent the number of logical processors (LP). LP is a term that is common for cloud instances to represent a vCPU. Recommended default setting is All LPs. In some cases, setting this option to Single LP can improve performance. Intel HT is only available in P-Core processors. Values for this BIOS option can be:

- **Single LPs:** run a single logical processor per core (same as one thread per core, or previously known as Hyper-Threading **Disable**)
- ***All LPs:** run two logical processors per core (same as two threads per core, or previously known as Hyper-Threading **ALL**)

A.15 Latency-Sensitive Recommendations

These recommendations are specifically provided for those interested in latency-sensitive system optimization, in other words, those primarily targeted for the Financial Services industry. These recommendations may also be applicable to other services where predictable performance with low jitter is desired.

- End users may be willing to trade off-peak performance for consistent performance.
- End users may not be as concerned about power savings.
- End users may be willing to consider trade-offs on long-term reliability for short-term performance gains.

A.15.1 Energy/Performance Bias Recommendation for Latency-Sensitive Applications

Energy Performance Bias should be set to "Performance" to optimize latency or "Balanced Performance" to weigh optimization toward performance while conserving energy. Checking to see whether this is the case can be slightly more complicated than one might think. In some cases, it may not be desirable for the operating system to control this feature, because it could adversely affect the performance-optimized experience. In addition, it is important that these settings be consistent on every thread across all cores.

The system can be set up by the BIOS to have the Energy Performance Bias operating mode controlled by the operating system (through the architectural MSR IA32_ENERGY_PERFORMANCE_BIAS at offset 0x1B0) or through another register that is intended to only be accessed by the BIOS.

- If the POWER_CTL MSR at offset 0x1FC bit [25] = 1, then the BIOS is in control of the energy performance bias.
 - In this case, check the ENERGY_PERF_BIAS_CONFIG MSR at offset 0xA01 bits [6:3].

- If the value is 0x0, then the system is in performance mode.
- If not, then it is not in performance mode, and a change is required in the BIOS to select the correct mode.
- If the POWER_CTL MSR at offset 0x1FC bit [25] = 0, then the operating system is in control of energy and performance bias determination.
 - In this case, select Performance mode from the OS (which should write 0x0 into IA32_ENERGY-PERFORMANCE_BIAS MSR 0x1B0 bits [3:0]).

Using Performance mode will configure the system to optimize for performance and latency and not power savings. Many of the features that are tuned for power savings will not benefit most latency-sensitive users which tend to disable the overall features (for example, control of the C-state and P-state algorithms). However, it will disable memory power management (Clock Enable [CKE] in particular).

A.15.2 Intel® Turbo Boost Technology Recommendation for Latency-Sensitive Applications

Disabling Intel® Turbo Boost Technology may achieve a more consistent performance without any negative impact on reliability. On the Intel® Xeon® E5 v4 family, and the Intel® Xeon® E7 v4 family, cores running Intel® AVX do not automatically decrease the maximum turbo frequency of other cores in the socket. On the Intel® Xeon® processor Silver, Intel® Xeon® Gold processor, and Intel® Xeon® Platinum processor Scalable families, cores running Intel® AVX2 or Intel® AVX-512 do not automatically decrease the maximum turbo frequency of other cores in the socket. Consequently, depending on how many cores are utilized, varying amounts of turbo can be achieved. On most processors, the highest turbo frequency is achieved when one or two cores are active.

Note: These per-core P-states are managed by the processor and are largely independent of the operating system. Specifically in performance mode, most operating systems will request the maximum frequency for each core, depending on how many cores are active and the overall power consumption of the processor. The actual frequency will be somewhere between the single-core turbo and the base frequency.

Per-core P-states also provide better power efficiency by allocating power only to the cores that need it, while inactive cores would drop to the minimum frequency of 1.2 GHz. All-core turbo frequencies are generally similar to previous generations.

When minimal performance jitter and variability is required, Intel recommends limiting or disabling the Intel® Turbo Boost Technology. TURBO_RATIO_LIMIT MSRs 0x1Ad [Ratios] and 0x1Ae [Cores] can be programmed to limit the amount of turbo upside that may exist for different core counts. See Section 7.1.17 Turbo Mode Overview in the *Birch Stream BIOS Writer's Guide* for an overview of the Intel® Turbo Boost Technology feature.

IA32_MISC_ENABLES MSR 0x1A0A.1.4 bit [38] can be set to one to completely disable turbo. This change results in a nearly constant performance, independent of how many cores are active, at the expense of peak performance. Another option for a milder tradeoff of performance for power is by redefining a single TRL bucket for any number of active cores running at P0n, the all core turbo frequency.

In addition to per-core turbo states, some of the Intel® Xeon® processor Scalable families also scale the uncore frequency independently of the cores. This feature provides further power efficiency gains but also introduces additional variability. On BHS platforms, a major performance-per-power tradeoff is the uncore frequency. In this technical article, uncore refers to the coherent fabrics and L3 caches in the compute chiplets. Setting the uncore frequency to its maximum can lead to significant power consumption, even at lower utilization levels (for example, 30%-50% utilized). On Intel Xeon 6 SoCs, each chiplet—compute and I/O— features separate fabrics that can be independently controlled. The new knobs give enhanced control over the uncore frequency policies.

Given these significant changes to the Intel® Turbo Boost Technology, Intel recommends customers to reevaluate the benefit of enabling turbo for their most critical workloads. Due to varying power characteristics, some parts with the Intel® Turbo Boost Technology 2.0 may not achieve maximum turbo frequencies when running certain workloads and using multiple cores concurrently. Availability and frequency upside of the Intel® Turbo Boost Technology 2.0 state depends on a number of factors including, but not limited to the following:

- Type of workload
- Number of active cores
- Estimated current consumption
- Estimated power consumption
- Processor temperature

When the processor is operating below these limits and the user's workload demands additional performance, the processor frequency will dynamically increase until the upper limit of the frequency is reached. The Intel® Turbo Boost Technology 2.0 has multiple algorithms operating in parallel to manage current, power, and temperature to maximize frequency and energy efficiency.

A.15.3 Enhanced Intel SpeedStep® Technology Recommendation for Latency-Sensitive Applications

The Enhanced Intel SpeedStep® Technology allows the processor to dynamically adjust frequency and voltage based on power versus performance needs.

Disabling Enhanced Intel SpeedStep® Technology in BIOS ensures the system will run at the processor base frequency (it will not use lower frequency states, see section 7 of the *Birch Stream BIOS Writer's Guide (BWG)*, document number 647898, for more information). Otherwise, Enhanced Intel SpeedStep® Technology must be enabled to utilize Intel® Turbo Boost Technology. Most operating systems today, when run in performance mode, will always request the maximum frequency, making the disabling of Enhanced Intel SpeedStep® Technology somewhat unnecessary.

Frequency transition can still occur for several reasons. Platform issues (for example, thermal throttling) will still cause a frequency transition, and disabling this behavior is not recommended.

Intel® architecture-based processors have several C-States giving a range of power savings and wake-up latencies. Keep in mind, also, that the behavior in a particular C-state on Intel® Xeon® Scalable processors may be different from C-states referred to with the same number on previous products (see *Section 7.5 CPU Power States (C-States) in the BWG* for more details). The greater the C-State, the more power is reduced (in other words, C6 is a lower power state than C1). When all cores request C1E, frequency will drop to the minimum frequency. This is independent of the Enhanced Intel SpeedStep® Technology configuration in BIOS (see MSR_POWER_CTL MSR 1FCh for more information).

A.15.4 BIOS Core Disable Recommendation for Latency-Sensitive Applications

It is possible to disable cores in BIOS (in other words, disable four of the cores on an eight-core processor) while retaining the full amount of L3 cache. For most workloads, the Intel® Turbo Boost Technology can be enabled and effectively get the maximum number of frequency upside bins for the reduced number of cores 100% of the time. The disabled cores are effectively forced into a deep power-saving state. This provides power which benefits Intel® Turbo Boost Technology. High power workloads that use Intel® AVX may not benefit. This can be a useful way to get a large cache, high frequency configuration.

A.15.5 Intel® Hyper-Threading (Intel HT) Technology Recommendation for Latency-Sensitive Applications

Intel does not usually recommend disabling Intel® HT Technology in the BIOS. However, the applications that are latency-sensitive may consider making this change.

Intel® HT Technology may decrease the performance of individual threads. This usually improves the overall system throughput. Many end users in the Financial Services Space are less interested in overall system throughput and are more concerned with individual thread throughput and performance stability. A subset of High-Performance Computing (HPC) workloads will degrade in performance when Intel® HT Technology is enabled and the cores use all available threads. Disabling Intel® HT Technology in BIOS is one option to avoid this problem. A second option is to spawn threads based on the number of cores available and not the number of threads available in the system. Each core is assigned one thread. As an example, in a 24-core processor where a program only uses 24 threads, the same performance is delivered whether Intel® HT Technology is enabled or disabled. The processor itself does not change its configuration in any way that impacts performance when disabling Intel® HT Technology. There are no queues or resources that are statically allocated across the threads when Intel® HT Technology is

enabled. If the second thread on a core is simply not used, then the performance of the core will match the case of disabling Intel® HT Technology on that core.

In latency-sensitive applications, where the additional threads are not needed, Intel® HT Technology should just be disabled in BIOS.

NOTE: The E-core processors (codename Sierra Forest) are single threaded and do not support Intel HT.

- Intel® Data Direct I/O Technology (Intel® DDIO)
Recommendation for Latency-Sensitive Applications

The Intel® Xeon® processor Scalable product family supports a feature called Intel® DDIO. This feature allows PCIe* devices to write data directly into a subset of the L3 cache. This can improve latencies for the core consuming data from the PCIe*. It also allows the PCIe* to read data directly out of the L3 cache rather than needing to go to memory. This feature should be enabled by default in BIOS, and it is recommended that this not be changed.

There is one x4 PCIe* port, Port 0, which does not support this feature. It is not uncommon for these four lanes to be incorporated into an x8 or x16 slot on boards, and in this case there is no functionality issue. It is recommended that this x4 port is not used for NICs that may benefit from this feature.

A.15.6 Restrictions to Micro-ops using *H instructions on Dispatch Ports

Some micro-architecture performance improvements required some restrictions to the micro-ops used for the *H¹ instructions (e.g., ADD, SUB, DEC, INC, NEG, XOR, MOV, MOVZX...). This decision was to ensure that the additional memory and integer dispatch ports added for performance would meet the timing, performance and area requirements within the core design specification. For the Golden Cove architecture, the *H operations are executed on Port 1 only, with 3 cycles latency. For more details, see the Intel® 64 and IA-32 Architectures Optimization Reference Manual, chapters 2.4 Golden Cove Microarchitecture Overview and 3.5.2.3 Partial Register Stalls, doc #821612.

¹ *H refers to partial registers like {AH, BH, CH, DH}

A.16 Benefits of Intel® Speed Select Technology (Intel® SST)

Intel Speed Select Technology is a feature that started with 2nd Gen Intel Xeon Scalable processors (those SKUs with a "Y" as part of the model number (example: Intel Xeon Gold 6336Y). There are four types of technology, performance profile (Intel SST-PP), base frequency (Intel SST-BF), core power (Intel SST-CP), turbo frequency (Intel SST-TF). Complete details on these features are described in external Intel SST documents.

An OS Mailbox Interface is available for all Intel SST features. The BIOS Mailbox interface is available for Intel SST-PP, Intel SST-BF, and Intel SST-CP,

where MSR PLATFORM_INFO bit[63] indicates availability of the Intel® SST-PP feature. SW reads this MSR and uses OS mailbox to discover the total # of Intel SST-PP levels. Refer to Intel® Speed Select Technology (Intel® SST) website for more information on Intel® SST frequency/performance modes. <https://www.intel.com/content/www/us/en/architecture-and-technology/speed-select-technology-article.html>. Three configuration profile levels are available and are referenced as Configuration 0 (PP0), configuration 3 (PP3), and configuration 4 (PP4).

A.16.1 Intel® SST – Performance Profile (Intel SST-PP)

The Intel Speed Select Technology – Performance Profile (Intel SST-PP) provides the capability to configure the CPU to run at one of several distinct profiles, with each profile differentiated by various performance vectors including the number of active cores, TDP, SSE base frequency, and Tjmax. OS/SW support and responsibility to maintain active core configurations can be managed out-of-band (OOB) or with hardware guided scheduling (HGS). Configuration parameters include: Core mask, Intel AVX2 and Intel AVX-512 base frequencies, turbo ratio limits for SSE/Intel AVX2, Intel AVX-512, uncore P0/P1, and memory frequency.

Performance of the 2nd and 3rd profiles will likely be better than or equal to comparable non-Intel SST processors of the same configuration due to additional LLC cache and power/thermal headroom.

Intel SST-PP allows the user to choose a performance profile level:

- *Auto:** Keep as HW selected SST-PP level
- Level 0:** SST-PP config level 0
- Level 1:** SST-PP config level 1
- Level 2:** SST-PP config level 2
- Level 3:** SST-PP config level 3
- Level 4:** SST-PP config level 4

Network CPUs (usually those suffixed with “N”), refer to Intel SST-PP profiles as:

- Network mode** – MODE1 (0) is optimized for 5G, Networking, and Edge Workloads, specifically high deterministic throughput, and low latency especially for packet processing
- Server mode** – MODE2 (1) is the same frequency/performance mode as general purpose CPUs used for Enterprise Apps
- Compute mode** – MODE3 (2) is optimized for maximum compute applications

A.16.2 Intel® SST – Base Frequency (Intel SST-BF)

With Intel Speed Select Technology-Base Frequency (Intel SST-BF), you can trade off base frequency between higher and lower priority cores. This could improve overall performance by giving the workload a few critical cores at a higher frequency.

Visit this link for configuration guidelines:

<https://builders.intel.com/docs/networkbuilders/intel-speed-select-technology-base-frequency-enhancing-performance.pdf>.

A.16.3 Intel® SST – Core Power (Intel SST-CP)

With Intel Speed Select Technology - Core Power (Intel SST-CP), the OS or virtual memory manager assigns a weight to each core. When surplus frequency is available, Power Control Unit (PCU) distributes surplus frequency according to cores' weights. Traditionally, PCU distributes surplus power equitably (also known as with no priority) amongst cores. Frequency prioritization works with SSE, Intel AVX2, and Intel AVX-512. With the Intel SST-CP features, overall performance could be improved by directing frequency to the highest priority or bottleneck cores. A core's priority can be changed at runtime via class of service (CLOS) interfaces. Starting with 3rd Gen Intel Xeon processors (codenamed Ice Lake), there were 4 defined CLOS buckets. Changing a CLOS profile instantly changes all cores associated with the profile.

There are two priority modes, ordered throttling priority and proportional priority.

Ordered: Higher priority cores get the excess budget first, are throttled last, and CLOS settings will impact low priority core performance.

Proportional: Available budget is shared among all cores based on configured priority weight after the minimum frequency requirements are met.

A.16.4 Active Turbo Frequency

All core active turbo frequency (P0n) is used as the maximum turbo frequency when all cores are active. Similarly, single core active turbo frequency (P01) is used as the maximum turbo frequency when a single core is active. For the purposes of determining maximum turbo frequency on the Intel® Xeon® Scalable processors, a core is considered active if it is not in C6.

The CPU has an inbuilt inertia for entering core C6 to compensate for the overhead of C6 entry and exit. Intel® Xeon® Scalable processors have increased inertia for entering core C6 due to the larger L2 cache. Workloads that run at P0n or P01 scenarios consistently should not observe any change due to this increased inertia. However, if a workload is run with additional applications that run in background and periodically wake up, they might keep the lightly loaded cores out of C6, thereby keeping the frequency for the active core from going above P0n.

B Recommended Memory Configurations for Best Performance

Next are the recommended memory configuration guidelines to optimize for the best application performance:

- Use identical DIMMs for all DIMM slots populated.
- Use dual rank registered DDR5 DIMMs at the fastest supported speed.
 - Dual rank and quad rank DIMMs will perform better than single rank DIMMs.
 - Single rank DIMMs will limit the performance of memory-intensive workloads, whether in one or two DIMM-per-channel configurations.
- Ensure all memory channels are populated equally.
 - At the same memory speed, two DIMMs per channel may perform slightly better than one DIMM per channel.
- DIMM size will depend on the application needs.

For applications or benchmarks that benefit from memory capacity more than memory performance, such as TPC-C* and TPC-E*, there are two options. Option one is likely best for TPC-C* since it allows a greater memory capacity.

1. Use the largest DIMMs available, currently expected to be 128 GB or 256 GB quad rank DDR5 LRDIMMs, and populate the maximum DIMMs allowed.
2. Use the largest dual rank registered DIMMs available. This is currently expected to be 64 GB dual rank DIMMs.

Populate as many channels per socket as possible prior to adding additional DIMMs to the channels. This will maximize the socket channel bandwidth available. Always use identical DIMMs in the system and ensure all memory channels are populated equally. Mixing of Error Correction Code (ECC) and non-ECC DIMMs is not supported.

B.1 Page Policy: (a.k.a. DRAM Page Mode)

A memory controller “page” is data in the DRAM row buffer. Open (adaptive) page mode keeps the page open for some amount of time after an access. Closed page mode closes the page immediately after access. Tradeoff between the lower latency of a page hit and the additional latency of a page miss (which requires a page close before opening a new one). Setting the memory controller page policy to closed may improve performance.

Values for this BIOS option (Socket Configuration > Memory Configuration > Page Policy > Page Policy) can be:

- **Open Adaptive Page:** use this setting for energy efficiency and server-side Java workloads or any workloads sensitive to idle memory latency.
- ***Closed Page:** use this setting for IO, OLTP and HPC applications.

Starting from BKC#122 (Granite Rapids-AP), BKC#110 (Granite Rapids-SP), IFWI 2024.47.2.04 (BIOSID 3544.P24), the BIOS default is changed from Open to Closed Page Mode.

The following WLs/KPIs would benefit from closed page mode:

- PCIe (SNC3): OOB Cluster Mode default
 - Multi-port BW (RdWr):
 - Closed Page policy shows upto 18% increase for cache-aligned payloads greater than 32B
 - Single-port BW (RdWr):
 - Closed Page policy shows upto 10% increase for 512B payloads
 - PCIe (HEX): AI HeadNode Configuration
 - Multi-port (eg 4x16, 5x16) BW: Closed Page policy shows upto 10% increase in BW for cache-aligned payloads greater than 128B
 - Single-port BW: no delta
 - DSA (MemMove, MemFill, MemCompare) :
 - With Closed Page Policy, BW improves upto 10% across the board both in HEX and SNC3
 - MLC BW Load-line.
 - At 25% load-line (or lower):
 - HEX: Open PP gives <2% BW for MLC W2, W3; no impact otherwise
-

- SNC: Closed PP improves BW across MLC patterns from **upto 11%** (11% for W6)
- At 50-75% load-line:
 - HEX: Closed PP improves W6 BW upto **5%** with little/no impact otherwise
 - SNC: Closed PP improves BW across MLC patterns from upto **13%** (13% for W6)
- At 100% load-line:
 - HEX: Closed PP improves W6 BW by **~2.5%** with no other impact
 - SNC: Closed PP improves BW across all MLC patterns from upto **9%** (9% for W6)

However, closed page mode increases memory sequential Read idle latency up to 20ns. As a result, selective Core performance benchmarks may be impacted (e.g. SIR 0.3%, Stream 1.5% and SpecJBB 2.3%). PCIe in SNC clustering mode small packet (8/16/32B) 100R/1R1W sequential access also shows upto 10% decrease in BW. Customers wanting to revert back to Open page mode can do this in BIOS knob.

See the *Intel® Xeon® Scalable family / Birch Stream Platform Memory Configuration Guide* for additional recommendations of configuration and memory slots options.

B.2 Non-Temporal Write Operations

Non-temporal writes (NTWs) can be a useful method to avoid cache pollution when writing to a location that is not expected to be reused in near future. NTWs may also reduce memory bandwidth requirements. Full line non-temporal writes do not require a read followed by write on the memory interface required for writes done through regular stores, avoiding a read-for-ownership operation followed by a writeback when the line is evicted from the cache. However, one of the drawbacks of NTW operations is that subsequent accesses after the write require the data to be fetched from memory since it is evicted out of the cache hierarchy. As a result, subsequent reads to those locations would see memory latency, whereas if a regular write was used instead of non-temporal writes, subsequent reads could hit in one of the cache hierarchy levels and complete much faster.

On 1st and 2nd Gen Intel Xeon processors, a per-core performance impact could be seen when a small number of threads (less than 6) were doing NTWs, whereas increasing the number of threads helped mitigated the lower bandwidth. NTW optimizations for 3rd and next Gen Intel Xeon Scalable processors have improved low core count active throughout. See the *Non-Temporal Write Optimizations for Intel Xeon Processor Scalable Family White Paper*, doc #636079, for more details.

Some of the potential usages and trade-offs with non-temporal writes are as follows:

- Access patterns that update large data structures not likely to be reused, cannot fit on the on-chip caches, or occupy a large portion of the on-chip cache replacing other useful data, can benefit from non-temporal writes. For example, STREAM benchmark for measuring memory bandwidth benefits from using non-temporal write.
 - Similarly, memset and memcpy functions that initialize or copy a block of memory can benefit from non-temporal writes if the block size is beyond a threshold where cache pollution can become a concern. For example, glibc library in Linux* uses non-temporal writes for memset and memcpy functions, but the behavior changes significantly between different versions of the library and the processor.
 - Some networking drivers and libraries may also use non-temporal writes for buffer copies; however, the benefit may depend on the size of the buffer being copied and how those buffers are used by the application subsequently.
 - Some of the non-volatile storage drivers may use non-temporal writes as a way to drain data from processor caches to persistent memory instead of using explicit cache line flush operations.
-

C Accelerator Performance Tuning

This section is an introduction about the new accelerators that were introduced in the 5th Gen Intel Xeon Scalable processor family.

C.1 Intel® Advanced Matrix eXtensions (Intel® AMX)

Intel® Advanced Matrix Extensions (Intel® AMX) is a 64-bit programming paradigm consisting of two components: A set of 2-dimensional registers (tiles) representing sub-arrays from a larger 2-dimensional memory image and an accelerator that can operate on tiles. The first implementation of this accelerator is called TMUL (tile matrix multiply unit) and was introduced in 4th Gen Intel Xeon processors.

- Intel AMX is available in p-core processors

Note: For more information about intrinsics, programming, and turbo frequencies, see section A.14.3 Recommended Performance Optimizations for Intel® Advanced Matrix Extensions (Intel® AMX). Visit ark.intel.com to confirm which Intel Xeon processors support Intel AMX.

C.2 Intel® QuickAssist Technology (Intel® QAT)

The Intel® QuickAssist Technology (Intel® QAT) is a hardware accelerator was first integrated in several 4th Gen Intel Xeon processors and improves performance across applications and platforms that have a need for symmetric encryption and authentication, asymmetric encryption, digital signatures, RSA, DH, and ECC, and lossless data compression.

- See the Intel QAT performance optimization guide for Intel® QuickAssist Technology, Doc #709209, for suggestions on the integration of the Intel QAT software with an application stack.

Numerous drivers exist for Intel QAT. Resources, technical collateral & application examples can be found here: [Intel QuickAssist Technology Developer pages](#).

To determine if your processor contains one or more Intel QAT devices, use the following Linux command to check the number of QAT devices on XCC CPUs:

```
#lspci |grep 4940
```

To check the number of QAT devices on MCC CPUs:

```
#lspci |grep 4942
```

C.3 Intel® In-memory Analytics Accelerator (Intel® IAA)

The Intel® In-Memory Analytics Accelerator (Intel® IAA) is a hardware accelerator that was first integrated in several 4th Gen Intel Xeon processors and targeted for very high throughput compression and decompression combined with primitive analytic functions.

These analytic functions are commonly used for filtering data during analytic query processing. It primarily targets applications such as big-data and in-memory analytic databases, as well as application-transparent usages such as memory page compression. Other operations such as data integrity functions (e.g., CRC64) are also supported. The device supports light-weight compression schemes such as zero-compression as well as heavier formats such as Huffman encoding and Deflate. For the Deflate format, it supports indexing the compressed stream for efficient random access. For a detailed discussion of optimization and tuning, see Intel® In-Memory Analytics Accelerator Architecture Specification, doc #721858.

- Intel IAA and Intel DSA share the same hardware/software and programming interface. The referenced documents describe the Intel IAA specific functionality and the minor differences in interface from the base Intel DSA specification. One should refer to the Intel Data Streaming Accelerator Architecture Specification for details on the common elements.

To determine if your processor contains one or more Intel IAA devices, use the following Linux command to check the number of IAA devices:

```
#lspci | grep 0cfe
```

C.4 Intel® Data Streaming Accelerator (Intel® DSA)

The Intel Data Streaming Accelerator (Intel® DSA) is a high-performance data copy and transformation accelerator that was first integrated in several 4th Gen Intel Xeon processors, targeted for optimizing streaming data movement and transformation operations common with applications for high-performance storage, networking, persistent memory, and various data processing applications.

Intel DSA replaces Intel® QuickData Technology, which is a part of Intel® I/O Acceleration Technology. Intel DSA provides higher overall system performance for data mover and transformation operations, while freeing up CPU cycles for higher level functions. Intel DSA hardware supports high performance data mover capability to/from volatile memory, persistent memory, memory-mapped I/O, and through a Non-Transparent Bridge (NTB) in the SoC to/from remote volatile and persistent memory on another node in a cluster. It provides a PCI Express* compatible programming interface to the operating system and can be controlled through a device driver. Shared Virtual Memory (SVM) and Intel® Scalable IOV is supported. For a detailed discussion of optimization and tuning, see Intel® Data Streaming Accelerator Architecture

Specification, doc #671116 and the Intel Data® Data Streaming Accelerator User Guide, [doc #759709](#).

- **Disabled:** Enables Intel DSA
- ***Enabled:** Disables Intel DSA

To determine if your processor contains one or more Intel DSA devices, use the following Linux command to check the number of DSA devices:

```
#lspci | grep 0b25
```

C.5 Intel® Dynamic Load Balancer (Intel® DLB)

The Intel® Dynamic Load Balancer (Intel® DLB) is an accelerator that was first integrated in several 4th Gen Intel Xeon processors to provide fully virtualized producer-to-consumer queuing. The Intel DLB supports high queuing rates, load balancing across consumers, multi-priority queuing arbitration, multiple scheduling types, and efficient queue notification. The Intel DLB appears to software as a PCIe device.

- The legacy name for the Intel® Dynamic Load Balancer (Intel® DLB) is the Hardware Queue Manager (HQM).

For a detailed discussion of optimization and tuning, see Intel® Dynamic Load Balancer 2.0 Programmer Guide, doc #613545.

To determine if your processor contains one or more Intel DLB devices, use the following Linux command to check the number of DLB devices:

```
#lspci | grep 2710
```

C.6 Intel® Volume Management Device (Intel® VMD) and Intel® Virtual RAID on CPU (Intel® VROC)

Intel VMD provides reliability, availability, and serviceability (RAS) features for data centric NVMe storage applications like surprise hot-plug and LED management. Intel VMD helps replace a bad NVMe SSD without having to reboot the system, locate the failed SSD with a blinking LED, and direct-assign a group of NVMe SSDs to a virtual machine for better performance.

Intel VROC provides integrated RAID solutions for NVMe and SATA devices without the cost and complexity of a host bus adapter (HBA). Intel VROC provides RAID data redundancy for both boot and data volumes in local storage. RAID 0/1/5/10 are supported on Windows and Linux OSes.

Both features were first integrated in several 4th Gen Intel Xeon processors to provide volume management and data redundancy for maximum system up-time and easy maintenance of local storage.

For a detailed discussion of Intel VMD and Intel VROC optimization and tuning, see Intel Virtual RAID on CPU (Intel VROC) Linux User Quick Start Guide, doc

#751470. Get documents, drivers and more for Intel VMD at:
<https://www.intel.com/content/www/us/en/collections/products/memory-storage/volume-management-device.html>.

C.7 Intel® Software Guard Extensions (Intel® SGX)

Intel SGX is a system of architectural enhancements defined to help protect application integrity and confidentiality of data, and to withstand certain SW and HW attacks. Intel SGX provides enhanced security protections for application data independent of operating system or hardware configuration and provides greater resistance against physical attacks. This feature provides fine grain data protection via application isolation in memory.

To learn more about Intel SGX, visit
<https://www.intel.com/content/www/us/en/developer/tools/software-guard-extensions/overview.html>.

C.8 Compute Express Link™ (CXL)

Compute Express Link is an alternate protocol that runs across the standard PCIe physical layer and improves accelerator performance via memory coherency and direct accelerator memory access. It reduces latency for memory load / store functions compared to similar PCIe 5.0 transactions. The Birch Stream platform supports up to four devices via Compute Express Link (CXL) 2.0.

More information can be found here: www.computeexpresslink.org.

D Recommendations for Energy Efficiency

D.1 Energy Efficiency Introduction

The 5th Gen Intel Xeon processors introduced an optimized power mode that provides automatic power savings for select accelerated workloads. This setting is default enabled on Intel Xeon 6 processors. To return to 5th Gen performance optimized behavior, a new latency optimized mode has been created. New platform and component in-band and out-of-band telemetry, called Intel® Platform Monitoring Technology (Intel® PMT), enables monitoring and control of electricity consumption across a number of components, including core power states, memory transactions, UPI links and I/O ports.

D.2 Suggested Memory Configuration

There are many factors that are used to help determine an optimal memory performance per watt configuration. Since bandwidth is set for a given channel, and two DIMM per channel configuration should only be used if it is required to meet the overall system memory requirements. Consequently, one DIMM per channel results in higher energy efficiency. Additionally, memory channels should be balanced (in other words, equal number of DIMMs per channel) to get the highest BW per DIMM.

D.3 Workload Energy Efficiency Configurations

The following table contains a list of select available roadmap SKUs. Each selection will affect both maximum and idle power draws as well as platform performance levels. Generally, higher energy efficiency is obtained by choosing higher core-count, higher power processors. SPECpower* scores can vary significantly depending on system ingredients such as: LAN, drives, Fan Speed Control, BIOS and OS power management settings. These scores are derived from testing on Intel Reference Platforms. Reference platforms are primarily used for validation and, therefore, are less energy efficient than commercial systems. However, the results are useful in making relative comparisons.

List of Select Available Standard Birch Stream CPU SKUs (with DDR5)

Process or TDP	Processor Part Number, Core Count, Base Frequency, LLC Cache Size	Nominal SPECpower Peak Performance (ssj_ops)	Peak Platform Power (W)	Nominal SPECpower Score (ssj_ops/W)
330W	Intel Xeon 6 (QDF: Q46C), 144C, 2.2GHz, 108MB (turbo ON) ¹	18,622	832	13,700
330W	Intel Xeon 6 (QDF: Q4WS), 144C, 2.2GHz, 108MB (turbo OFF) ¹	21,636	842	16,790

500W	Intel Xeon 6 6980P (QDF: Q5SM), 128C, 2.0GHz, 504 MB ²	33,146,654	1,771	13,051
------	---	------------	-------	--------

Notes:

System Under Test – Hardware Configuration:

- Motherboard: BHS 2S commercial baseboard Base S7Q
- ¹ RAM: 1024GB (16x 64GB DDR5-6400) 1 DPC
- ² RAM: 1536GB (24x 64GB DDR5-6400) 1 DPC
- 1x storage device (NVME)
- 2x nnnW Titanium PSU
- Fans:
- # of Sockets: 2 Socket system
- Turbo setting: Enabled

System Under Test – Software Configuration:

SPECpower* ssj2008-v1. Xx

Recommended OS: Use Linux distribution for publication work.

OS Power Profile: Power Saver (with modifications on next slide)

OS Large Pages: Enable large pages policy for the user running the workload (i.e. Administrator).

Set to power saver mode power profile, then modify as below.

Check current setting (Default is 0)

powercfg /q scheme_current sub_processor perfboostmode

**Power Setting GUID: be337238-0d82-4146-a960-4f3749d470c7
(Processor performance boost mode)**

GUID Alias: PERFBOOSTMODE

Possible Setting Index: 000

Possible Setting Friendly Name: Disabled

Possible Setting Index: 001

Possible Setting Friendly Name: Enabled

Possible Setting Index: 002

Possible Setting Friendly Name: Aggressive

Possible Setting Index: 003

Possible Setting Friendly Name: Efficient Enabled

Possible Setting Index: 004

Possible Setting Friendly Name: Efficient Aggressive

Possible Setting Index: 005

Possible Setting Friendly Name: Aggressive At Guaranteed

Possible Setting Index: 006

Possible Setting Friendly Name: Efficient Aggressive At Guaranteed

Current AC Power Setting Index: 0x00000000

Current DC Power Setting Index: 0x00000000

Set New Value For Turbo Boost Mode

**powercfg /setacvalueindex scheme_current sub_processor
perfboostmode 3**

Double check perfboostmode has changed from 0 to 3

powercfg /qh scheme_current sub_processor perfboostmode

Current AC Power Setting Index: 0x00000003

Current DC Power Setting Index: 0x00000000

powercfg /setactive scheme_current

Reboot System

Recommended JDK: Use JDK 17 for your publication work.

Optimal Java options to use for JDK 17

BKM Affinity settings are two hardware threads per JVM instance (one physical and its logical twin).

```
-server -Xmn1825m -Xms1995m -Xmx1995m -XX:SurvivorRatio=1 -  
XX:TargetSurvivorRatio=99 -XX:ParallelGCThreads=2 -  
XX:AllocatePrefetchDistance=256 -XX:AllocatePrefetchLines=4 -  
XX:LoopUnrollLimit=45 -XX:InitialTenuringThreshold=12 -  
XX:MaxTenuringThreshold=15 -XX:InlineSmallCode=3900 -  
XX:MaxInlineSize=270 -XX:FreqInlineSize=2500 -XX:+UseLargePages -  
XX:+UseParallelGC -XX:UseAVX=0 -XX:BiasedLockingStartupDelay=30000 -  
XX:-UseAdaptiveSizePolicy -XX:+UseBiasedLocking -XX:+UseOptoBiasInlining -  
XX:+UseFastStosb -XX:MaxInlineLevel=9 -XX:MaxVectorSize=32 -XX:-  
UseSharedSpaces -XX:-UseXMMForObjInit -XX:-UseBASE64Intrinsics -  
XX:+OptimizeFill -XX:+CriticalJNINatives
```

One of the recommended changes as of June 2024 is a change in the heap size from -Xmn1300m -Xms1550m -Xmx1550m to -Xmn1825m -Xms1995m -

Xmx1995m. The result varies from sku to sku but the heap size is slightly more optimized, and it tends to produce a 7/10th of 1% to 1% improvement in the overall score due to improved power. This is more noticeable with Turbo enabled.

Set affinity with the following patterns:

```
start /node 0 /affinity 3 cmd /k %JAVA% %JAVAOPTIONS_SSJ%  
org.spec.power.ssj.SpecPowerSsj -jvmId 1 -numJvms %JVMS% -director  
%DIRECTOR_HOST% -setid %SETID%
```

```
start /node 0 /affinity C cmd /k %JAVA% %JAVAOPTIONS_SSJ% org.spec.power.ssj.SpecPowerSsj -jvmId 2 -  
numJvms %JVMS% -director %DIRECTOR_HOST% -setid %SETID%
```

```
start /node 0 /affinity 30 cmd /k %JAVA% %JAVAOPTIONS_SSJ% org.spec.power.ssj.SpecPowerSsj -jvmId 3 -  
numJvms %JVMS% -director %DIRECTOR_HOST% -setid %SETID%
```

```
start /node 0 /affinity C0 cmd /k %JAVA% %JAVAOPTIONS_SSJ% org.spec.power.ssj.SpecPowerSsj -jvmId 4 -  
numJvms %JVMS% -director %DIRECTOR_HOST% -setid %SETID%
```

```
start /node 0 /affinity 300 cmd /k %JAVA% %JAVAOPTIONS_SSJ% org.spec.power.ssj.SpecPowerSsj -jvmId 5 -  
numJvms %JVMS% -director %DIRECTOR_HOST% -setid %SETID%
```

```
start /node 0 /affinity C00 cmd /k %JAVA% %JAVAOPTIONS_SSJ% org.spec.power.ssj.SpecPowerSsj -jvmId 6 -  
numJvms %JVMS% -director %DIRECTOR_HOST% -setid %SETID%
```

...

```
start /node 3 /affinity 3000000000000000 cmd /k %JAVA% %JAVAOPTIONS_SSJ%  
org.spec.power.ssj.SpecPowerSsj -jvmId 123 -numJvms %JVMS% -director %DIRECTOR_HOST% -setid  
%SETID%
```

```
start /node 3 /affinity C000000000000000 cmd /k %JAVA% %JAVAOPTIONS_SSJ%  
org.spec.power.ssj.SpecPowerSsj -jvmId 124 -numJvms %JVMS% -director %DIRECTOR_HOST% -setid  
%SETID%
```

```
start /node 3 /affinity 3000000000000000 cmd /k %JAVA% %JAVAOPTIONS_SSJ%  
org.spec.power.ssj.SpecPowerSsj -jvmId 125 -numJvms %JVMS% -director %DIRECTOR_HOST% -setid  
%SETID%
```

```
start /node 3 /affinity C000000000000000 cmd /k %JAVA% %JAVAOPTIONS_SSJ%  
org.spec.power.ssj.SpecPowerSsj -jvmId 126 -numJvms %JVMS% -director %DIRECTOR_HOST% -setid  
%SETID%
```

```
start /node 3 /affinity 3000000000000000 cmd /k %JAVA% %JAVAOPTIONS_SSJ%  
org.spec.power.ssj.SpecPowerSsj -jvmId 127 -numJvms %JVMS% -director %DIRECTOR_HOST% -setid  
%SETID%
```

```
start /node 3 /affinity C000000000000000 cmd /k %JAVA% %JAVAOPTIONS_SSJ%  
org.spec.power.ssj.SpecPowerSsj -jvmId 128 -numJvms %JVMS% -director %DIRECTOR_HOST% -setid  
%SETID%
```

Optimal Bios settings are:

Hardware Prefetcher: Disabled

Adjacent Prefetcher: Disabled

DCU Streamer: Enabled

DCU IP Prefetcher: Enabled

Virtual Numa Enabled

LLC Deadline alloc: Enabled

Opportunistic LLC to SF Migration: Disabled

Hardware P-State: Disabled

C-states C6: No Limit

Turbo on/off needs to be tested for each sku to determine which produces the best overall score.

Note: The results published in this section are merely demonstrative and do not reflect the best energy efficiency/performance these or any other units can achieve.

D.4 Networking Workload Energy Efficiency Considerations

Network transformation is fueled by 5G deployment to move fixed function network application to be software defined allowing more flexibility to the Communication Service Providers deployment. Even so, some mission critical network functions are still required to have determinism expected performance with extremely low latency. Some network functions can tolerate some latency and low jitters allowing consideration for energy efficiency. The associated tuning guides cover many BIOS options that can be considered for performance tuning and optimization. For networking, there are 3 workload profiles that target different workload tolerance to jitter and latency. The following are the 3 workload profiles:

- The Network Function Virtualization Infrastructure (NFVI) profile targeting the Network Function Virtualization Infrastructure that may be able to tolerate p-state and c-state transitions taking full advantage of the Energy-Balance with Turbo Boost technology

For a detailed discussion of optimization and tuning, see BIOS Settings for Intel® Wireline, Cable, Wireless and Converged Access Platform Section 3, Doc #747130.

- The Network Function Virtualization Forwarding Platform (NFVI FP) profile is defined for intensive Data Plane requirement for deterministic performance and able to tolerate some low latency jitters.

For a detailed discussion of optimization and tuning, see BIOS Settings for Intel® Wireline, Cable, Wireless and Converged Access Platform Section 2, Doc #747130.

- The virtualized Radio Area Network (vRAN) profile is defined for Intel FlexRAN™ Software. This is the most conservative and stringent BIOS profiles here to ensure high performance for AVX512, 5G ISA instructions and extremely low latency requirement.
-

For a detailed discussion of optimization and tuning, see BIOS Settings for FlexRAN™ Reference Architecture Platforms based on Intel® Xeon® processors Section 2.3 and Section 3, Doc #640685.

E Firmware and BIOS Settings Guidelines for Energy Efficiency

The following guidelines may be used when configuring the BIOS for optimum energy efficiency. These guidelines are applicable to server class platforms and are tailored to give the optimum power performance for Birch Stream 2S server-based products.

First, ensure that you have the latest platform firmware installed. Current firmware revision levels can be obtained from your Intel FAE. Next, verify that all capabilities to support processor deep sleep states (C6) and Closed Loop Thermal Throttling (CLTT), if applicable, are enabled. To improve energy efficiency and TCO). Finally, use the following guidelines to ensure that all power management features are properly configured.

Note: These suggested settings are only guidelines for SPECPower* type applications. The BIOS you are working with may have different or missing settings. Several iterations of BIOS settings may be necessary to obtain optimum performance and/or power efficiency.

Birch Stream BIOS Settings Guidelines for Energy Efficiency

Processor Configuration		Comments
Intel® HT Technology (ALL)	Enabled (default)	
DCU IP Prefetcher	Enabled (default)	
Hardware Prefetcher	Disabled	
Adjacent Cache Prefetch	Disabled	
LLC Prefetch	Disabled	
DCU Streamer Prefetcher	Disabled	
LLC Dead Line Allocation	Enabled	
Socket Configuration		
PCI-E ASPM Support	L1 Only (default)	
Advanced Power Management Configuration		
Intel® Turbo Boost Technology	Disabled (SKU dependent)	Some SKUs may provide better energy efficiency with turbo disabled
CPU C State	Auto (default)	CPU C States supported and enabled are C0, (Fast) C1E and C6.
CPU C6 Report	Auto (default)	
Package C State Limit	Auto (default)	CPU Package C States supported is PC0, PC2 and PC6 (C6-P)
Hardware P-States	OOB	Default is Native Mode

Processor Configuration		Comments
EPP Profile	Power	
Latency Optimized Mode	Enabled	Default disabled to bring better performance per Watt across the load line.
Energy Perf BIAS		
Power Performance Tuning	Enable (default) BIOS Controlled	When choosing OOB / Power mode, the values for this will be fixed and unchangeable by the end user. It will be greyed out.
Energy_Perf_Bias_Cfg Mode	Balanced Performance (default)	When choosing OOB / Power mode, the values for this will be fixed and unchangeable by the end user. It will be greyed out.
Common RefCode Configuration		
NUMA	Enabled (default)	
Intel® UPI Configuration (Uncore Configuration)		
Link Speed Mode	Fast (Default)	
Link Speed Frequency	Auto	
SNC	Enabled	Sets two cluster mode which is dependent upon both this setting being enabled and Max Rank Interleaving in iMC set to one way.
Memory Configuration		
Max Rank Interleaving in iMC	One way	
Memory Power & Thermal Configuration (under Advanced Power Management Configuration)		
Set Throttling Mode	CLTT (default)	
Memory RAS Configuration		
Patrol Scrub	Disabled	BIOS default is *Enable at End of POST
IIO Configuration		
Wake On LAN from S5	Disabled (default)	
USB Configuration		
xHCI Mode	Enable (default)	
PCH Configuration		
PCH DM1 ASPM	ASPM L1	
PCIe* Root Port Function Swapping	Disable	
PCH DMI ASPM Testing	ASPM L1	

E.1 Power-Related BIOS Knobs

Legacy software is common in the data center. Due to the cost and risk associated with software updates, new server platforms are frequently coupled with 4–6-year-old Operating Systems (OS). These legacy operating systems are not enabled or tuned for the latest power management technologies, which can result in lower-than-expected performance or higher than expected power. Hardware Power Management (HWPM) addresses this issue by embedding frequency control capabilities previously owned by operating systems, in CPU firmware. The firmware is optimized for the target CPU micro-architecture and feature set, ensuring data center operators can realize the full benefit of new power management technologies at launch.

HWPM also addresses the need to support a broad range of power and performance requirements - enabling data center operators to dynamically switch between different “profiles” optimized for highest performance, lowest power, or a balance between the two. Historically, adoption of power management features is limited when there are no clear mechanisms to tune or control feature behavior.

E.2 Hardware P-States (HWP)

Hardware P-states (HWP) refers to the set of algorithms used to dynamically determine core frequency. By fixing the core frequency to P1, HWP is disabled. Hardware P-states (HWP) are the key components of HWPM which deals with P-state control. HWP works by freeing the operating system from making direct frequency decisions. HWP supports two modes: HWP Out-of-Band (OOB) mode and HWP Native mode.

- ***Native Mode:** the OS provides high level hints or constraints to hardware and hardware makes pstate decisions by observing core statistics and OS-provided guidance or hints.
- **Out-of-Band (OOB):** hardware autonomously makes pstate decisions with OS completely excluded from this process.
- **Disable:** hardware uses legacy pstate MSRs to determine desired core frequency which do not support autonomous pstate decisions.
- **Native Mode with No Legacy Support:** Same as native mode except HWP is enabled by BIOS instead of OS.

E.3 Energy Perf Bias Cfg Mode

Energy Performance Bias (EPB) determines whether the pcode algorithms are biased more towards performance (at the expense of power) or power (at the expense of performance). Balanced Performance and Balanced Power are in-between settings that try to balance power and performance.

- **Performance:** biases the algorithms toward maximum core performance and memory bandwidth.
- ***Balanced Performance:** slightly bias performance over energy savings.
- **Balanced Power:** slightly bias energy savings over performance.
- **Power:** Biases the algorithms towards saving power.

E.4 Dynamic Loadline Switch

When cores have a high turbo frequency residency, Energy Perf Bias (EPB) will autonomously change to performance so that power management will prioritize performance over power savings. When average core p-state residency drops below turbo, EPB will return to original values.

- ***Enable:** Enable the loadline switch
- **Disable:** Disable the loadline switch

E.5 ACPI C6x Enumeration

The actions for C6 state include core clock gating, core cache flush, core in reset, and core FIVR/PLL shutdown. C6 essentially powers off the core. Beginning with 6th Gen Intel Xeon Scalable family, support for different MWAIT encodings for Core C6 and Package C6 is exposed in BIOS.

- ***Auto:** Enable Core C6 and Package C6
- **Disable:** Disable Core C6, disable package C6
- **C6 as ACPI C2/C3:** Enable core C6, disable package C6
- **C6-P as ACPI C2/C3:** Enable core C6, enable package C6

E.6 C1 to C1e Promotion

On top of C1 action of core clock gating, the C1E enhanced halt state adds voltage-frequency reduction to Pm frequency. Beginning with 4th Gen Intel Xeon Scalable family, fast and autonomous per-core C1E states are implemented. Enabled by default, core voltage and core frequency are reduced for energy efficiency, while having very low exit latency and high responsiveness, improving reliability for users who typically disable C6. C1 to C1e promotion will treat all C1 requests as C1e requests. This is to support OS that have not enabled C1e MWAIT requests.

- ***Enable:** Enable Core C1E autopromotion from C1
- **Disable:** Disable Core C1E autopromotion from C1

E.7 Uncore Freq Ratio (COMPUTE|IO)

Uncore Frequency Ratio refers to the algorithms used to dynamically determine uncore frequency. By setting the uncore frequency to 0, uncore frequency will dynamically change based on utilization. If setting to a value other than 0, autonomous uncore frequency decisions are disabled and will run at the specific

fixed frequency. Compute and IO tiles have independent controls, and the valid range of fixed frequency values is listed in uncore ratio range for each tile type.

- ***[0]:** Enable uncore frequency scaling algorithms
- **[X]:** Disable uncore frequency scaling algorithms and run at fixed frequency X

E.8 Uncore Freq Control (COMPUTE|IO)

RAPL (Running Average Power Limit) algorithm limits socket power by reducing core and uncore frequencies. In this power limiting case, the uncore frequency will reduce proportionally with the core frequency given by this UFS RAPL line equation: $\text{UNCORE_RAPL_LIMIT} = \text{UFS_RAPL_BASE} + \text{CORE_RAPL_LIMIT} * \text{UFS_RAPL_SLOPE}$.

- ***Mode 1:** Enable frequency selection based on UFS RAPL line in the power limiting case
- **Mode 0:** Uncore frequency will not reduce in power limiting cases

E.9 Latency Optimized Mode

The Birch Stream platform now has a new performance-per-Watt biased power mode that is "default" along with a lower latency mode called "latency optimized mode".

- ***Disable:** Out-of-box mode for newer disaggregated SoC arch with significant power savings across the load line. Better performance per watt across the load line. No significant difference in performance compared to Latency Optimized Mode at 100% Load Level.
- **Enable:** Uncore frequencies will run up to their maximum limits within the RAPL budget. This mode is not performance-per-watt optimized across the load line.

E.10 Intel® Speed Shift Technology

The Birch Stream server processor supports the Intel® Speed Shift Technology, where the OS can provide hints to the PCU via an MSR regarding the desired power/performance.

The PCU can use these hints to manipulate the internal behavior of the processor and determine whether to target higher performance or better power savings.

The user selects a particular power policy within the OS (for example, performance, balanced or power-save in Windows*). Based on this setting and other factors, the OS decides to send the appropriate policy request for each logical processor via the Hardware-Controlled Performance States / Hardware P-states (HWP) MSR. The OS input is provided to the hardware via the

Energy_Performance_Preference field (bits 31:24) of IA32_HWP_REQUEST (MSR 774h). This provides a continuous scale from 0 to 255 to choose from, with 00h, 7Fh, BFh, and FFh thresholds corresponding to the following energy policy profiles: Performance, Balanced Performance (recommended default), Balanced Energy, and Energy Efficient profile, respectively.

Core Frequency typically depends on CPU utilization. A lower EPP setting causes the system to ramp up frequency more aggressively to minimize system response time even at low CPU utilization. On the other hand, a high EPP setting results in lower frequency being used at low CPU utilization to achieve higher energy efficiency.

Note: Intel recommends that customers evaluate their respective workloads to choose a suitable energy policy profile that meets their need.

Energy Performance Bias (EPB)

Feature	Performance	Balanced Performance	Balanced Energy	Energy Efficient
Turbo	Aggressive		Clipped	
Memory CKE	OFF	Moderate	More Aggressive	
Memory OSR	OFF		Not Aggressive	
Intel® UPI L0p	Moderate		More Aggressive	
C-state Demotion	Normal	Normal	Normal	Normal
I/O BW P-Limit	Normal	Normal	Normal	Normal

The Intel® Speed Shift Technology supports an OOB mode that could be enabled in BIOS. In this mode, power scheme selection is done entirely under hardware control without input from OS. A BIOS knob is used to select one of the four power schemes previously mentioned; however, this knob is currently not operational. Nonetheless, power scheme selection could be done by updating bits 31:24 of IA32_HWP_REQUEST (MSR 774h) for all logical processors.

Note: Disabling dynamic switching and running the system in any mode except performance will result in turbo not granted.

If turbo is disabled, core frequencies will not get to the turbo region (in other words, the OS will not ask for turbo). Maximum performance will be limited to what is achievable with the P1 frequency. However, performance benefits from actions listed under Performance Mode in the previous table can be realized by setting Operating System Power Management (OSPM) to performance.

The I/O BW P-Limit is a feature that monitors the Intel® UPI, memory BW and PCIe* bandwidth; and uncore frequency is adjusted when certain thresholds are crossed.

F Other Considerations

F.1 VR Design

Optimal system performance/watt will be heavily weighted on how well your voltage converters perform the energy transfer function or in other words, how efficient each of your platforms on-board voltage regulators are in doing their job.

Voltage Regulator Conversion Efficiency (η) is defined as the ratio of Output Power (P_{out} W) to Input Power (P_{in} W), whereas the difference of the two is Power Loss (P_{loss} W), which is the amount of wasted power (heat).

$$\eta(\%) = \frac{P_{out}(W)}{P_{in}(W)} * 100 \quad P_{loss}(W) = P_{in}(W) - P_{out}(W)$$

F.2 Efficiency and Power Loss Equations

There are many forcing functions that contribute to how efficient a voltage converter design will be and the designer should start with an understanding of the primary contributors to Power Loss. The remaining portion of this section provides a brief description of each power loss contributor.

Note: The description is focused purely on conversion efficiency improvements. The pros and cons of each are not discussed and are outside the scope of this section.

$$P_{switching}(W) + P_{conduction}(W) + P_{pcb}(W) + P_{inductor}(W) + P_{pwm \& driver}(W) = P_{loss}(W)$$

F.3 Elements that Contribute to Power Loss

Converter topology: Example linear versus switching regulators, linear will typically have poorer conversion efficiency than switchers for an equivalent static load. This is especially so for higher current required converters.

Input voltage to output voltage difference: For a switching converter, the smaller the difference from input voltage to output voltage (in other words, the longer the ON time of the high-side MOSFET) the more efficient the conversion will be. In general, and for example, a 12V to 1.2V switching converter will be less efficient than a 9V to 1.2V switching converter for an equivalent static load.

Switching losses: The lower the switching frequency, the more efficient the switching voltage converter will be. Simply stated, there are more electrical switches due to the higher switching frequency and therefore the higher number of switching losses there are for an equivalent time sample. Switching

losses can be reduced by increasing the MOSFET's gate drive strength, lowering the converter's settable switching frequency and using a smaller MOSFET. To help understand the elements that contribute to switching losses in a basic buck voltage regulator, here is the switching losses formula:

$$P_{\text{switching}}(W) \cong \frac{1}{2} * V_{\text{in}}(V) * I_L(A) * (T_{\text{rise}} + T_{\text{fall}}) * F_{\text{switching}}$$

Where:

$V_{\text{in}}(V)$ = Input Voltage,

$I_L(A)$ = Average Output Inductor Current,

T_{rise} = High-Side MOSFET rise ON time,

T_{fall} = High-Side MOSFET fall OFF time and

$F_{\text{switching}}$ = the converter's set switching frequency

F.4 MOSFET and Freewheeling Diode Conduction Losses

In a 12V to 1.2V switching converter with a static load, the low-side MOSFET will be enhanced (ON or conducting) 90% of the time for a given period. With regards to conversion efficiency decisions and in selecting a low-side MOSFET, designers should focus on the conduction resistance, R_{DSon} (OHM) in the MOSFET components datasheet. The lower the RDS-On value, the lower the conduction losses will be. The MOSFET conduction loss equals to $I_{\text{mosfet}}(A)^2 R_{\text{DSon}}(\text{OHM})$. Similarly, conduction losses exist in the freewheeling diode, $V_{\text{diode}}(V) * I(A)$.

F.5 Printed Circuit Board (PCB) Losses

PCB power delivery fill's size, layer count and Cu weight will represent I^2R power losses. This can be viewed as a resistive path and thus the lower the resistive path, the lower the voltage drop and power losses. The greater each of these are the lower the PCB losses.

F.6 Inductor Losses

The windings of an output inductor will have some Direct Current Resistance (DCR). The lower the selected output inductor's DCR, the lower the inductor losses.

F.7 Controller and Driver Losses

IC losses are the product of the input voltage multiplied by the Quiescent current. The IC Quiescent current will be provided in the vendor's datasheet. Additionally, dead time is defined as the time when the low-side and high-side MOSFETs are both OFF. This dead time should be kept to the absolute minimum and in doing so will reduce MOSFET driver losses. It is essential not to have the high-side MOSFET and low-side MOSFETs ON at the same time. Doing so would cause a dead short resulting in possible flames. There are many MOSFET drivers that have implemented dynamic dead time reduction circuits to help minimize this time.

Another factor that should be considered is the removal of "lossy" test hooks. Test hooks can contribute to lower voltage converter's efficiency. An example of this is validation "Sense Resistors" that are used to measure output current of a voltage converter. Removing of all non-essential or unused test hooks for productized boards is highly recommended.

F.8 Motherboard VR13.0 6 Phase Shedding in Low Power Mode

The VR 13.0 controllers are available with auto phase shedding to continuously select the right number of phases and optimize for efficiency over the entire load range.

F.9 VCCIN MBVR IMON (Iout) Accuracy versus System Performance

Unlike in earlier platforms, MBVR IMON accuracy is a platform design choice for platforms. Customers are free to buy "better" VRs if that is what they need to meet their design goals.

The Intel Xeon 6 processor PCU regulates directly to socket power as reported over MBVR IMON. No "adjustment factor is applied". Slope / offset registers are provided for adjustment if customers want, default is slope =1, offset =0 (no adjustment).

If MBVR consistently under-reports power, thermal throttling (or need for additional cooling) may result.

Average processor package power, as measured by VCCIN MBVR, is capped at the TDP limit. If the MBVR consistently over-reports power, CPU core-frequency product may be reduced by up to 2 MHz-core per "watt of over-reporting."

Example: Over reporting by 5W could result in 100 MHz lower turbo frequency for a high core count processor resulting in several percent performance loss depending upon application.

F.10 System Thermal Design Considerations

Employing solid design principles in system thermal/mechanical design and applying proper techniques to fine-tune thermal control systems will have a material effect on the overall energy efficiency of the platform. From a physical design perspective, the following areas should be considered:

- Minimize system impedance – improving airflow through the system will allow for the use of lower power/speed fans to be used to achieve the desired thermal performance. Venting/ free area ratio at the front and rear of the system enclosure should be maximized, particularly to improve airflow for core components (processor and memory). Care should also be taken to ensure cable routing is efficient and does not impede airflow to critical devices.
- Design ducting to minimize airflow bypass – most server systems employ ducting to direct airflow through heatsinks for high power dissipating devices. Techniques should be used to force the airflow through the heatsink fins, not allowing it to bypass to the side or over the top of the heatsink.
- Eliminate recirculation and minimize preheat – emphasis should be placed on reducing the amount of air recirculation within air ducts and minimizing pre-heating effects for core area components due to the power dissipation of upstream devices, such as disk drives.
- Heatsink selection – selecting a higher performance heatsink, with improved component to ambient heat transfer capability, lowers system airflow requirements. This, in turn, will allow for lower fan power to be applied to the design and improve overall energy efficiency.

In tuning thermal performance, the system designer needs to carefully balance the fan power applied to the design against the thermal requirements of the key power dissipating devices. Often, designers focus on the thermal performance needed for high system ambient conditions and high utilization rates and fail to properly tune the design at idle and low/nominal ambient and low utilization rates. For techniques in turning system fan speed control implementations, see the white paper: *[Romley, Grantley] Platform – Server System Thermal Design Considerations for Optimizing Energy Efficiency*, document number 472400.

For maximum power efficiency, all closed loop thermal throttling parameters should be enabled in both BIOS and by the operating system. All CPU and platform fans should have multi-speed capability. Low power acoustic settings are generally labeled “Quiet” or “Silent” in BIOS or OS options; “Performance” and “Max Speed” settings should be avoided.

F.11 Alternating Current (AC)/ Direct Current (DC) Power Supply Selection

Power supply (PSU) should be selected in accordance with:

- Section 3.1 of the white paper *Workstation and Server Platform Readiness Checks for the ENERGY STAR Program*, document number 411729;
- Appropriate section of the *ENERGY STAR Specification for Computers* (section 1.0 for Servers), and;
- *Purley Platform Power Budget Calculator*, document number 568840;
- *Common Redundant Power Supply Design Guide for Servers Common Redundant Power Supply Design Guide for Servers*, document number 573090;
- 80PLUS® – Certified Power Supplies site:
 - <http://www.plugloadsolutions.com/80PlusPowerSupplies.aspx>; and
 - http://en.wikipedia.org/wiki/80_Plus.

PSU Efficiency Level Certifications

80 Plus Test Type ▼	115V Internal Non-Redundant				230V Internal Redundant			
Percentage of Rated Load ►	10%	20%	50%	100%	10%	20%	50%	100%
80 Plus		80%	80%	80%				
80 Plus Bronze		82%	85%	82%		81%	85%	81%
80 Plus Silver		85%	88%	85%		85%	89%	85%
80 Plus Gold		87%	90%	87%		88%	92%	88%
80 Plus Platinum		90%	92%	89%		90%	94%	91%
80 Plus Titanium	90%	92%	94%	90%	90%	94%	96%	91%

F.12 USB Device Interactions

USB selective suspend –some peripherals do not support selective suspend, which can significantly impact the idle power due to poor PC6 residency.

The Windows* tool *powercfg* may provide suggestions on which devices are getting selective suspend.

Enhanced Host Controller Interface (EHCI) and Extensible Host Controller Interface (XHCI)

Generally, better PC6 residency is attained with XHCI.

- USB3 XHCI Scheduling is done in Hardware compared to scheduling being done in Software for EHCI.
- Periodic memory scans are eliminated for XHCI scheduling.

- XHCI reduces host memory references.



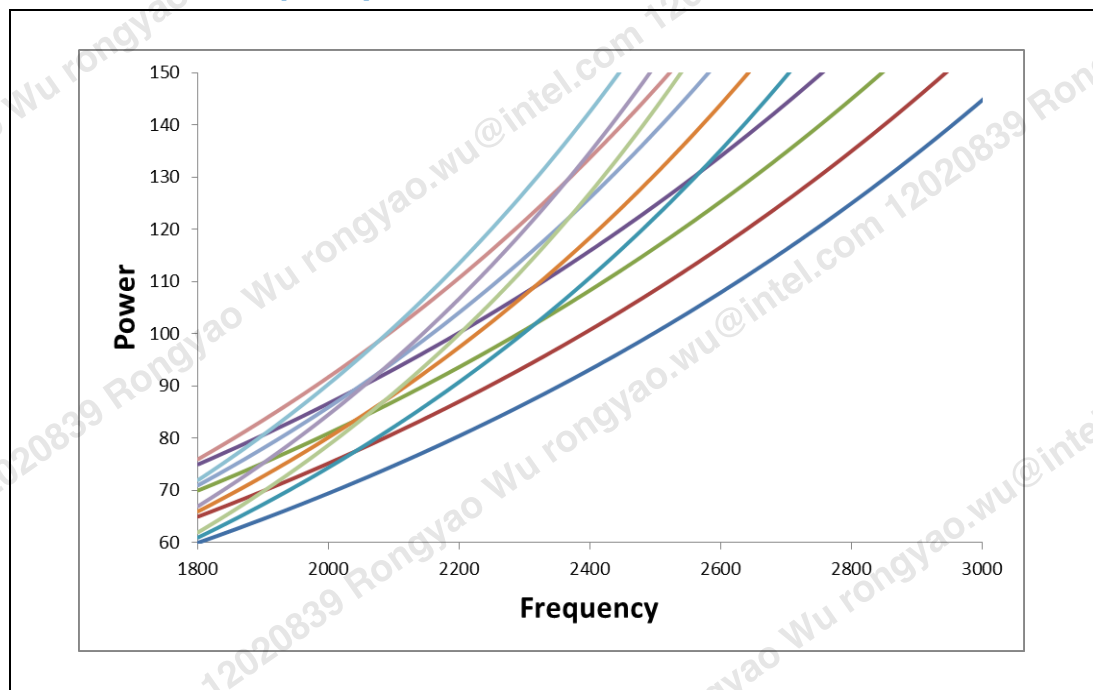
F.13 Processor Power Variation

In every manufacturing process, there is some level of measurable variation. This is certainly true for silicon devices. In the case of a CPU, the maximum frequency is limited to the speed of the slowest transistors, and because there are now over a billion of them in server processors, tiny amounts of variation have a huge impact. Power variation is determined by frequency of the processor, the voltage needed to drive the processor to that frequency, and the number of functional cores among other things.

The easiest way to visualize this is to think of a chart with two axes: Frequency and power at some fixed voltage and core count. Each curve represents a population of processors whose power increases with frequency.

Note: These curves are for the figure below only and are completely synthetic (made up) and do not represent any actual silicon.

Example of Processor Frequency and Power Distribution



Given such a distribution, it is simply not feasible to market and sell each processor according to its unique properties. So, like every industry with a similar variation in product capability, Intel groups similar parts using differentiating parameters that customers care about. Today these are primarily performance and performance per watt.

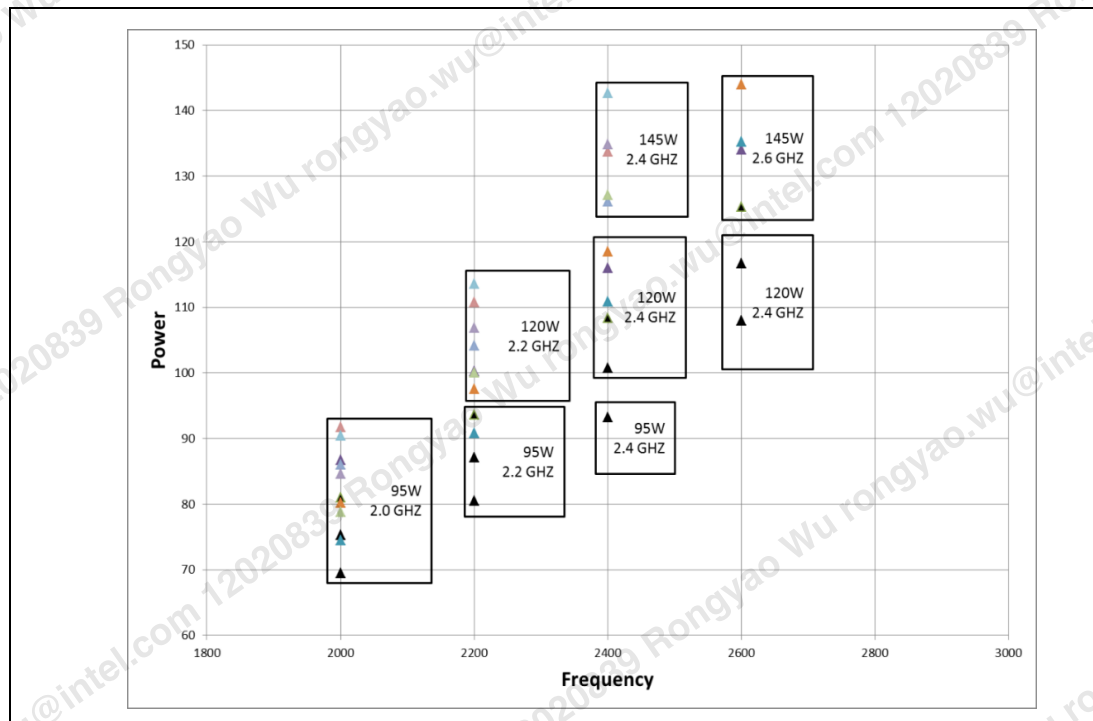
Because of this, when Intel creates a SKU stack, customers will commonly see frequency in approximately 100 or 200 MHz increments. Processor power distribution is less linear. For low power parts (<20W) customers, may see power increments of just a few watts. For high power parts (>100W), the difference between bins is commonly 10W or higher.

That leads to something that looks like the figure below.

Note: The chart is for the figure below only and is entirely synthetic and does not represent any actual silicon or binning strategies.

Because Intel customers demand the same performance (frequency) for every part in a SKU and desire each processor to use the lowest power possible to achieve that frequency, Intel does the following: Each processor is tested to find the maximum frequency bin that the part can fall in, and then its core voltages are set it to the lowest voltage (power) level that it can operate at and still maintain that bin frequency. In other words, on a part-by-part basis, Intel optimizes for performance per watt.

Example of Processor Frequency and Power Distribution by Bin



What this means is that the vast majority of processors at a particular frequency and core count will consume power that is somewhere between the TDP specified for that SKU and the TDP for the SKU below it. This assumes all parts are at the same sustained TDP work-load, and in the same environment.

Because of this, a common question is: "Can Intel provide the power distribution of a particular SKU?"

Unfortunately, it is not possible for a couple of reasons. Intel will commonly change up its binning strategies throughout the life of a product to meet customer demand. For example, processors capable of higher frequencies may be placed in lower frequency lower power bins. Intel also works constantly to improve our manufacturing process resulting in a general overall reduction in power for a given frequency. These two factors will cause the distribution of power to vary significantly over time. Therefore, Intel cannot provide a description of a SKU's power distribution or even the SKU's average processor power consumption and guarantee that distribution or average.

Glossary

Related Acronyms

Terminology/Acronym	Description
#	This symbol after a signal name means an active low signal (logic low = asserted).
Active High	Signal is asserted when logic is high or one.
Active Low	Signal is asserted when logic is low or zero volts.
ASPM	Active State Power Management.
AVP	Adaptive Voltage positioning or 339999 (LL).
BHS	Server platform (codename Birch Stream) using either Sierra Forest or Granite Rapids processor
BJT	Bipolar Junction Transistor.
CCM	Continuous Conduction Mode.
CFM	Cubic feet per minute (airflow).
CMRR	Common-Mode Rejection Ratio.
CPU	Central Processing Unit, microprocessor.
CRC-8	8-bit Cyclic Redundancy Check.
Alert#	SVID bus alert line, as known as SVID_ALERT# or interrupt line.
DAC	Digital to Analog Converter.
DCM	Discontinuous Conduction Mode.
DCR	Direct Current Resistance.
DCU	Data Cache Unit
D-VID	Dynamic Voltage Identification. A mode of operation where the output voltage is dynamically changed by changing the VID bits.
D-VID code	Dynamic Voltage Identification code.
EHCI	Enhanced Host Controller Interface.
EDS	Electrical Design Specification, processor specification document.
EGS	Server platform (codename Eagle Stream) using the 4 th or 5 th Gen Xeon processors.
EMI	Electromagnetic Interference.
EMR	Emerald Rapids
ESD	Electrostatic Discharge.
ESL	Equivalent Series Inductance.
ESR	Equivalent Series Resistance.
FET	Field Effect Transistor.
FMB	Flexible Motherboard.
FR4	A type of Printed Circuit Board (PCB) material.

Terminology/Acronym**Description**

GND	Ground (return).
HFM	High Frequency Mode.
HS	Heatsink.
HVM	High Volume Manufacturing.
IC	Integrated Circuit.
IIN (ICCIN)	Load current associated to the VIN (VCCIN) voltage rail.
Intel® UPI	Intel® Ultra Path Interconnect (Intel® UPI).
ITT	Bus current associated with the VTT (VCCIO) supply.
Iout	Analog output (0 to 900 mV) proportional to average output current.
IMON	Digitized version of Iout signal, available over SVID bus.
ISA	Load current associated to the VSA voltage rail.
KTI	See Intel® UPI.
LFM	Depending on context: linear feet per minute (airflow) or Low Frequency Mode.
Load Line	A mathematical model that describes voltage current relationship given system impedance (R_{LL}). The nominal load line equation is $VIN (VCCIN) = VID - I \cdot R_{LL}$.
LV	Low Voltage.
MCC	Medium core count die and contains up to 'n' cores depending on the processor family. For SPR, the processor die is monolithic up to 32 cores.
MBVR	Motherboard Voltage Regulator.
MLCC	Multilayer Ceramic Capacitor.
Monotonically	A waveform changes from one level to another in a steady fashion, without intermediate retracement or oscillation.
MOSFET	Metal Oxide Semiconductor Field Effect Transistor.
MTBF	Mean Time Between Failures.
OCP	Overcurrent Protection.
OTP	Over-Temperature Protection.
OVP	Overvoltage Protection.
Purley	Server platform codename using the SKL processor.
PEC	Packet Error Correction.
PM	Power Management.
POL	Point of Load.
PROCHOT#	Under thermal monitoring, the VRD asserts this processor input to indicate an over temperature condition has occurred. Assertion of this signal places the processor in a low power state, thereby cooling the voltage regulator.
PSD	P-State Dependency
PSI#	Power status indicator.
PWM	Pulse width modulation.
PWR_IN	Voltage regulator's input power (typically 12V).
RDSon	MOSFET's drain to source resistance.
Ripple AND Noise	The periodic or random signals measured over a frequency band of 0 Hz to 20 MHz.

Terminology/Acronym

Description

Rise time	Rise time is defined as the time it takes any output voltage to rise from 10% to 90% of its nominal voltage.
R _{LL}	Load line impedance. Defined as the nominal voltage droop/current ratio. This is the load line slope.
RMS	Root Mean Square.
RSS	Root Sum of Squares. A method of adding statistical variables.
SetWP	Working point is a new command used to move multiple rails on the SVID bus to a new VID with the broadcast or all call command address.
SFF	Small Form Factor.
Slope	Load line resistance. See R _{LL} .
SOC	System on Chip, IE CPU + PCH or other dies on 1 package, microserver CPUs SOC packages.
SPR	Sapphire Rapids
SVID	Serial Voltage Identification.
TDC	Thermal Design Current, aka Icc_TDC or Isa_TDC, refers to the continuous current the voltage rail (VIN or VSA) must be capable of delivering for an indefinite period of time.
TDE	Thermal Design Envelope.
TDP	Processor's total Thermal Design Power.
Thermal monitor	A feature of the voltage regulator that places the processor in a low power state when critical VRD temperatures are reached, thereby reducing power and VRD temperature.
Tolerance budget (TOB)	Defines the voltage regulator's 3-σ voltage variation across temperature, manufacturing variation, and age factors. Must be ensured by design through component selection.
Transient Load Line or AC LL	Equal to dV/dI or Vdroop/Istep and is controlled by switching frequency, decoupling capacitor selection and motherboard layout parasitic.
ULV	Ultralow Voltage.
UPI	Ultra path interconnect (4 UPIs for XCC, 3 UPIs for MCC)
UVLO	Undervoltage lockout.
Vcc	PWM chip's power rail.
VIN or VCCIN	Processor core voltage defined in the processor datasheet.
VMCP	Processor's MultiChip Package voltage.
XCC	Extreme core count die. For SPR, the processor die contains 4 tiles of up to 15 cores in each tile.
XHCI	Extensible Host Controller Interface.